Renal Physiology

Physiology of the Four Key Processes of the Kidney

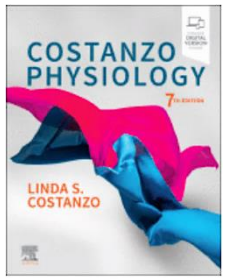
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Fall 2026

Learning Objectives

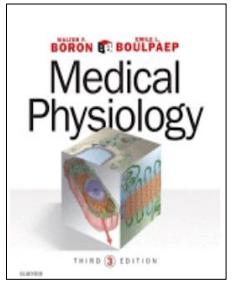
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2. Evaluate how the glomerular filtration barrier, Starling forces, and intrinsic autoregulatory mechanisms determine net filtration pressure and glomerular filtration rate.
3. Characterize the general epithelial transport mechanisms of proximal tubular reabsorption and peritubular secretion, highlighting their opposing roles in determining net excretion.
4. Describe how differing transport properties of the descending and ascending limbs of the Loop of Henle establish the hyperosmotic medullary gradient.
5. Explain how final urinary excretion and concentration are modulated in the collecting duct by the presence or absence of ADH and aldosterone.

Resources

READ



Chapter 6



Chapters 33 & 34

DO

* **TopHat Questions**
- **Guyton and Hall Review Questions (Library)**
- **Costanzo Cases**

TOP HAT

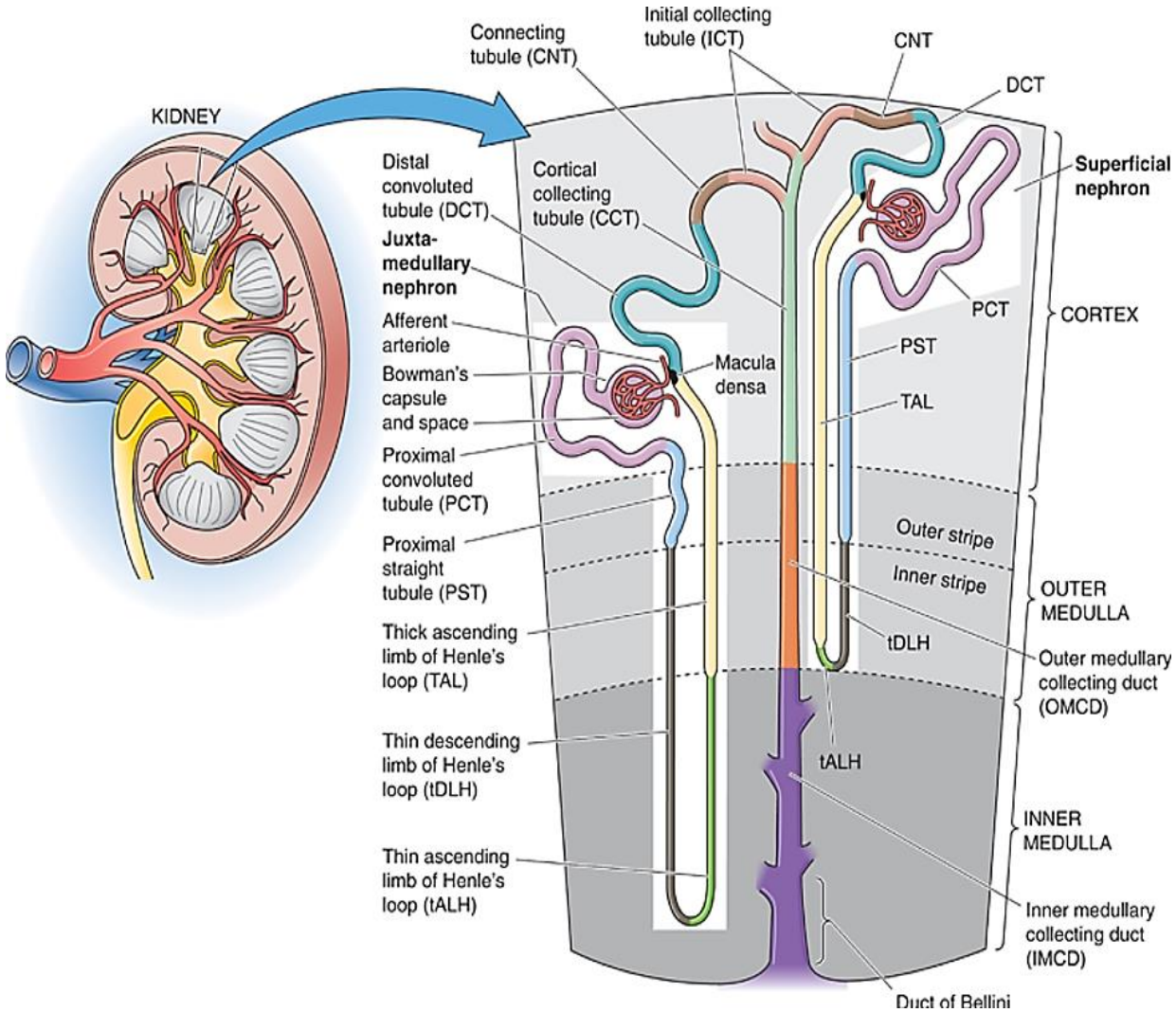
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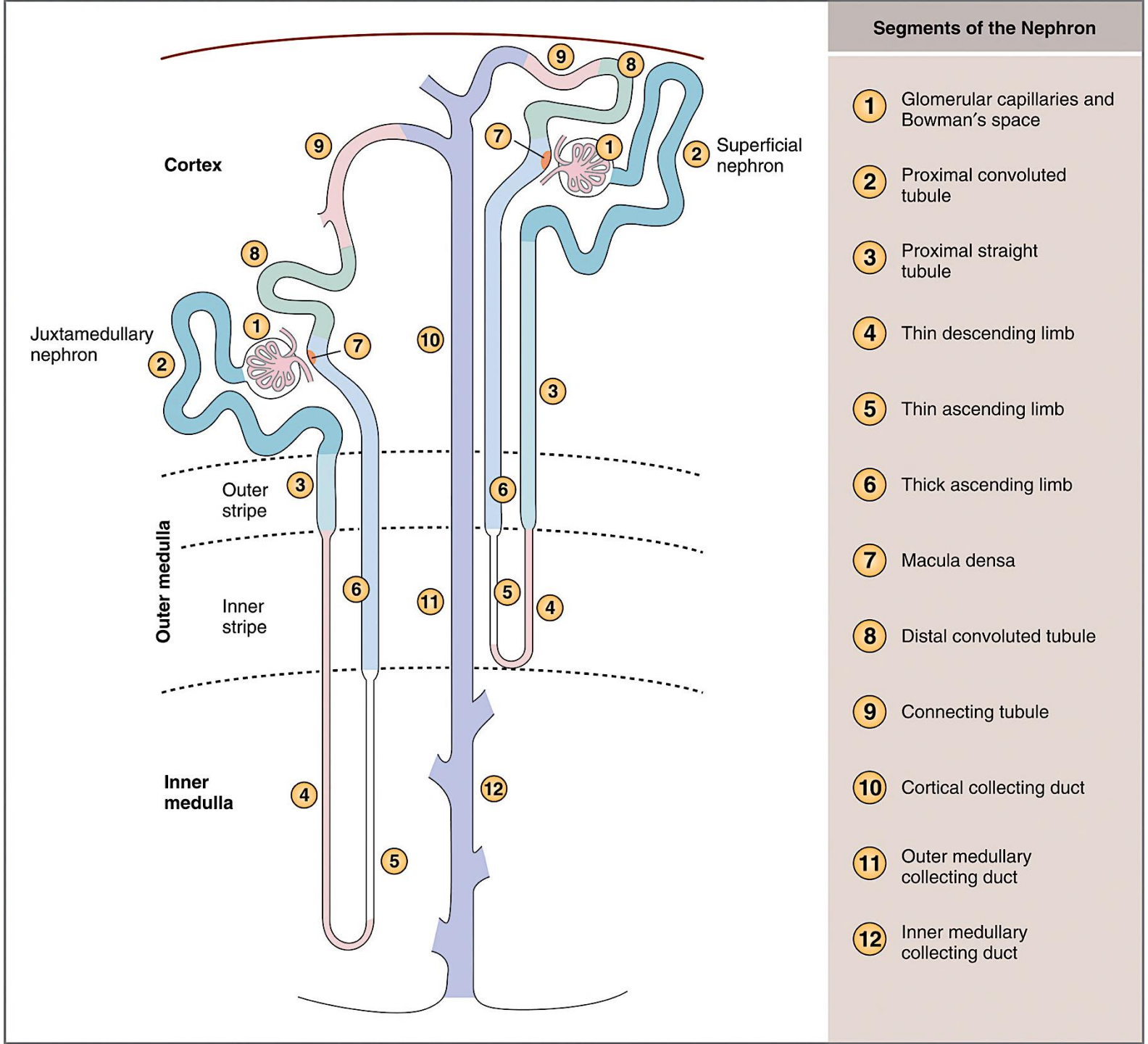
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The Nephron

- **Functional Unit of the Kidney:** Each kidney contains approximately 1 millionnephrons acting in parallel to process plasma.
- **Two Interdependent Components:**
  - *Vascular component:* Delivers blood under tightly regulated pressures to and from tubular transport sites.
  - *Tubular component:* A continuous hollow epithelial tube transforming glomerular ultrafiltrate into final urine.
- **Spatial Organization:** Glomeruli, proximal tubules, and distal convolutions reside exclusively in the renal cortex; the loops of Henle and collecting ducts extend deep into the renal medulla.

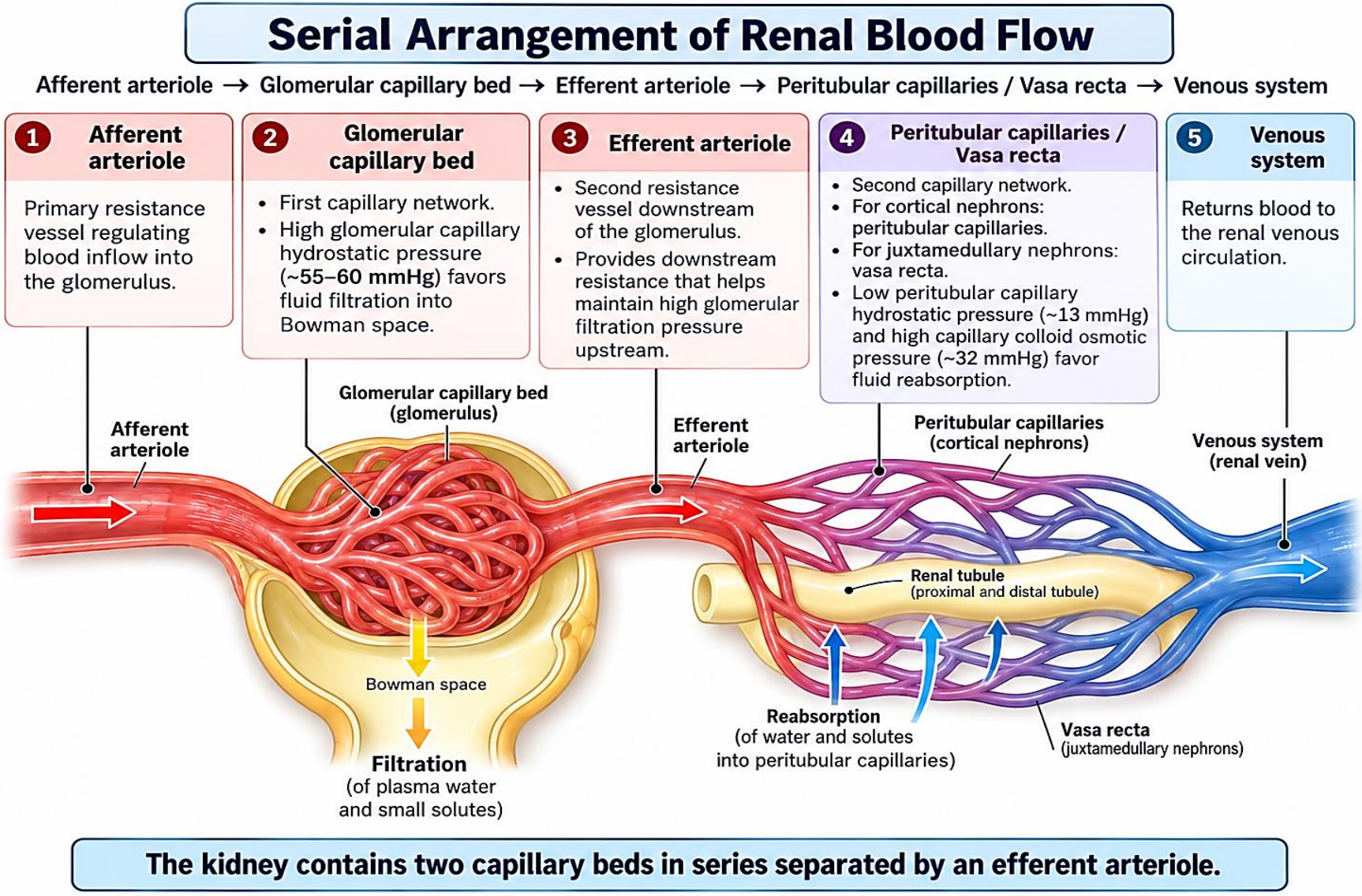



Cortex
Juxtamedullary nephron
Superficial nephron
Outer stripe
Inner stripe
Outer medulla
Inner medulla
Segments of the Nephron
1 Glomerular capillaries and Bowman's space
2 Proximal convoluted tubule
3 Proximal straight tubule
4 Thin descending limb
5 Thin ascending limb
6 Thick ascending limb
7 Macula densa
8 Distal convoluted tubule
9 Connecting tubule
10 Cortical collecting duct
11 Outer medullary collecting duct
12 Inner medullary collecting duct

Renal Portal System

- **A Portal System in Series:** Unlike typical systemic circuits (arteriole →single capillary bed →venule), the kidney routes blood through **two capillary beds in series** separated by an arteriole.
- **Pressure Tuning for Opposite Functions:**
  - *Glomerular capillaries:* Placed between two resistance arterioles (afferent and efferent) to maintain high hydrostatic pressure optimized for ultrafiltration.
  - *Peritubular capillaries:* Downstream of efferent resistance; hydrostatic pressure drops sharply while oncotic pressure remains elevated, creating ideal Starling forces to reabsorb fluid.
- **Vasa Recta Specialization:** Juxtamedullary efferent arterioles form long, unbranched hairpin loops descending alongside the loops of Henle into the medulla.

Blood Flow and Portal System



Renal Hemodynamics: Converting RBF to RPF

- **Renal Blood Flow (RBF):** The kidneys receive approximately $20–25\%$ of resting cardiac output (~ 1.1 L/min or 1100 mL/min) despite accounting for only 0.5% of body weight.
- **Why Focus on Renal Plasma Flow (RPF)?**
  - The cellular components of blood are normally excluded by the filtration barrier and ignore tubular transport processes.
  - The nephron filters, processes, and clears the **aqueous plasma**, not erythrocytes.
- **Calculating Renal Plasma Flow:**
  - $\text{RPF} = \text{RBF} \times (1 - \text{HCT})$
- **Standard Clinical Baseline:**
  - $\text{RPF} = 1100 \text{ mL/min} \times (1 - 0.43) \approx 625 \text{ mL plasma/min}$

Whole blood after centrifugation



Plasma fraction
≈ 55–60% of whole blood volume

- Water (≈ 90%)
- Plasma proteins (albumin, globulins, fibrinogen)
- Dissolved solutes (electrolytes, nutrients, hormones, wastes, etc.)

Buffy coat (very thin layer)
WBCs + platelets

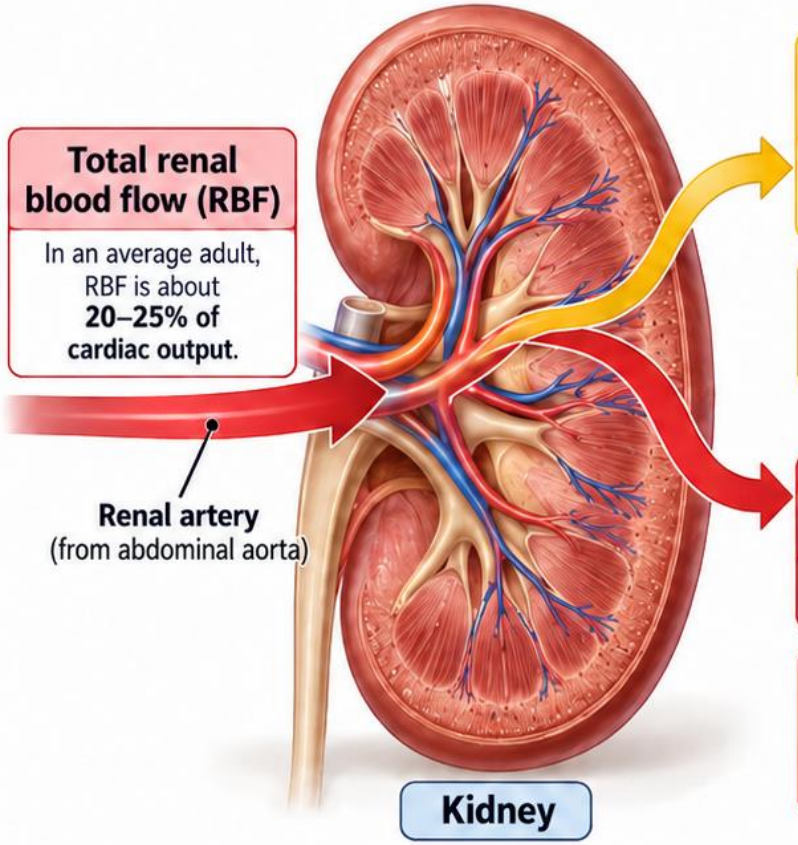
Hematocrit (HCT)
≈ 40–45% of whole blood volume

Fraction of whole-blood volume occupied by erythrocytes (RBCs).

Renal blood flow and its components

Total renal blood flow (RBF)

In an average adult, RBF is about **20–25% of cardiac output.**



Renal plasma flow (RPF)
= RBF × (1 – HCT)

≈ 55–60% of RBF
(because HCT ≈ 40–45%)

Plasma is the portion of renal blood flow available for glomerular filtration.

Cellular / erythrocyte flow
= RBF × HCT

≈ 40–45% of RBF

Erythrocytes (and other formed elements) remain within the renal vasculature and are **not filtered** at the glomerulus.

Total blood volume (composition)

Cellular elements (formed elements)
≈ 40–45%



Plasma
≈ 55–60%

- Water
- Proteins (albumin, globulins, fibrinogen)
- Dissolved solutes (electrolytes, nutrients, hormones, wastes, etc.)

Key takeaways

- **Hematocrit (HCT) ≈ 40–45%** → fraction of whole-blood volume occupied by erythrocytes (mainly RBCs).
- **Plasma fraction ≈ 55–60%** → water, proteins, and dissolved solutes.
- **Renal plasma flow (RPF) = RBF × (1 – HCT)** → portion available for glomerular filtration.
- **Cellular / erythrocyte flow = RBF × HCT** → erythrocytes remain in the renal vasculature and are not filtered.
- In an average adult, renal blood flow is about **20–25%** of cardiac output.

A 45-year-old male undergoes a physiological evaluation of renal function. Laboratory tests determine that his total Renal Blood Flow (RBF) is 1200 mL/min and his hematocrit (HCT) is 45% (0.45). What is this patient's Renal Plasma Flow (RPF)?

- **Identify the Relationship:** Whole blood consists of cellular elements and fluid plasma. Because cellular elements are excluded from ultrafiltration, the nephron acts upon the plasma fraction, defined as $(1 - \text{HCT})$.

- **Formula:** $\text{RPF} = \text{RBF} \times (1 - \text{HCT})$

- **Step-by-Step Calculation:**
  - $\text{RPF} = 1200 \text{ mL/min} \times (1 - 0.45)$
  - $\text{RPF} = 1200 \text{ mL/min} \times 0.55$
  - $\text{RPF} = 660 \text{ mL plasma/min}$

- **Physiological Takeaway:** Although 1200 mL of whole blood perfuses the kidneys every minute, only 660 mL/min of aqueous plasma is available to enter the renal portal microcirculation and undergo filtration and tubular processing.

The Fundamental Renal Mass Balance Equation

- **The Mass Balance Formula:**
  - Excretion $=$ Filtration $-$ Reabsorption $+$ Secretion
  - $E = F - R + S$
- **Filtration Fraction (FF):**
  - The proportion of renal plasma flow filtered into Bowman's space.
  - $\text{FF} = \frac{\text{GFR}}{\text{RPF}} = \frac{125\ \text{mL/min}}{625\ \text{mL/min}} = 0.20\ (20\%)$
  - The remaining 80% passes directly into the efferent arteriole and peritubular capillaries.
- **Scale of Daily Fluid Turnover (Water Baseline):**
  - *Filtration:* 125 mL/min (~ 180 L/day)
  - *Reabsorption:* 124 mL/min (~ 178.5 L/day, $> 99\%$ reclaimed)
  - *Excretion:* 1 mL/min (~ 1.5 L/day, $< 1\%$ lost)
- **Core Takeaway:** Every solute and water molecule's appearance in urine is governed strictly by the arithmetic sum of these four processes.

During a renal microperfusion study, the transport rates of an investigational drug are measured. The drug is freely filtered at the glomerulus with a filtration rate (F) of 150 mg/min. Proximal tubular reabsorption (R) reclaims 40 mg/min back into the peritubular capillaries, while active organic transporter secretion (S) adds 15 mg/min from the peritubular capillaries into the tubular lumen. What is the net urinary excretion rate (E) of this drug?

- **Identify the Relationship:** The final urinary appearance of any substance is determined by the renal mass balance equation: filtration minus reabsorption plus secretion.
- **Formula:** $E = F - R + S$
- **Step-by-Step Calculation:**
- $E = 150$ mg/min $- 40$ mg/min $+ 15$ mg/min
- $E = 110$ mg/min $+ 15$ mg/min
- $E = 125$ mg/min
- **Physiological Takeaway:** Because reabsorption (40 mg/min) exceeds secretion (15 mg/min), the net tubular effect is reabsorption (25 mg/min net reabsorbed), resulting in an excretion rate that is lower than the initial filtered load ($E < F$).

Four Key Processes of the Kidney

- So the “four major renal processes” are not just a list, they are the terms in a mass-balance relationship!

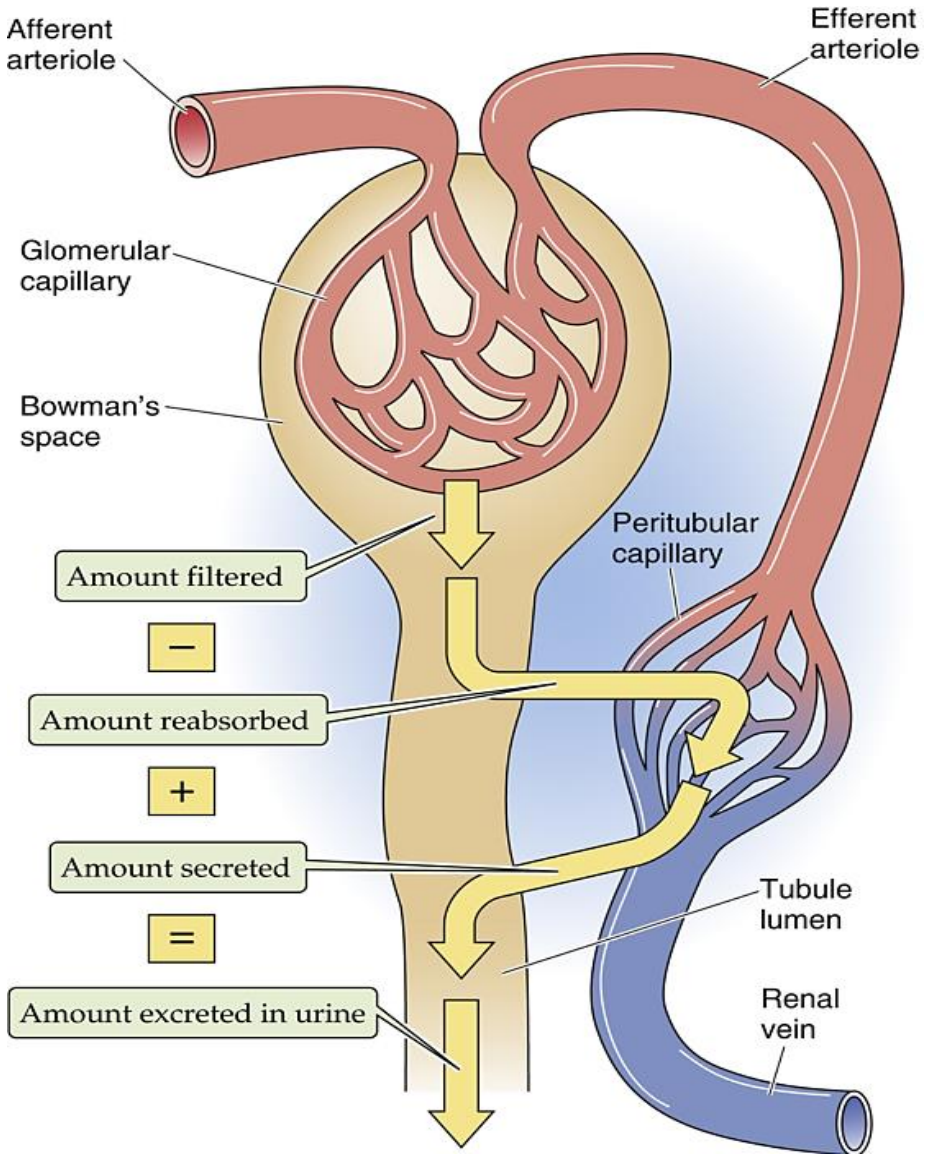
- **Filtration** adds X to tubular fluid.
- **Reabsorption** removes X from tubular fluid.
- **Secretion** adds more X to tubular fluid.
- **Excretion** is the net amount of X that leaves the body in urine.

- And because these are **rates**, you can write: $E_X = F_X - R_X + S_X$
  - Values are per unit time

Four Key Processes of the Kidney

- Filtration
- Reabsorption
- Secretion
- Excretion

- E = F + S – R



Interim Summary 1

- **Portal System in Series:** Sandwiching the glomerular capillaries between two muscular arterioles (afferent and efferent) maintains high hydrostatic pressure (P_{GC}) for filtration, while efferent resistance drops downstream peritubular hydrostatic pressure (P_{pc}) to favor reabsorption.
- **Aqueous Focus (**RPF**):** Red blood cells ignore nephron transport. We convert whole blood flow (RBF ≈ 1100 mL/min) to plasma flow ($\text{RPF} = \text{RBF} \times [1 - \text{HCT}] \approx 625$ mL/min) because the kidneys clear the aqueous compartment.
- **The Filtration Fraction (**FF $= 20\%$**):** Normal GFRis 125 mL/min (~ 180 L/day). Because urinary excretion is only 1 mL/min (~ 1.5 L/day), $> 99\%$of filtered water and solutes must be reabsorbed.
- **Governing Mass Balance:** Net excretion rate is strictly the algebraic sum of the three upstream rates: $E = F - R + S$.

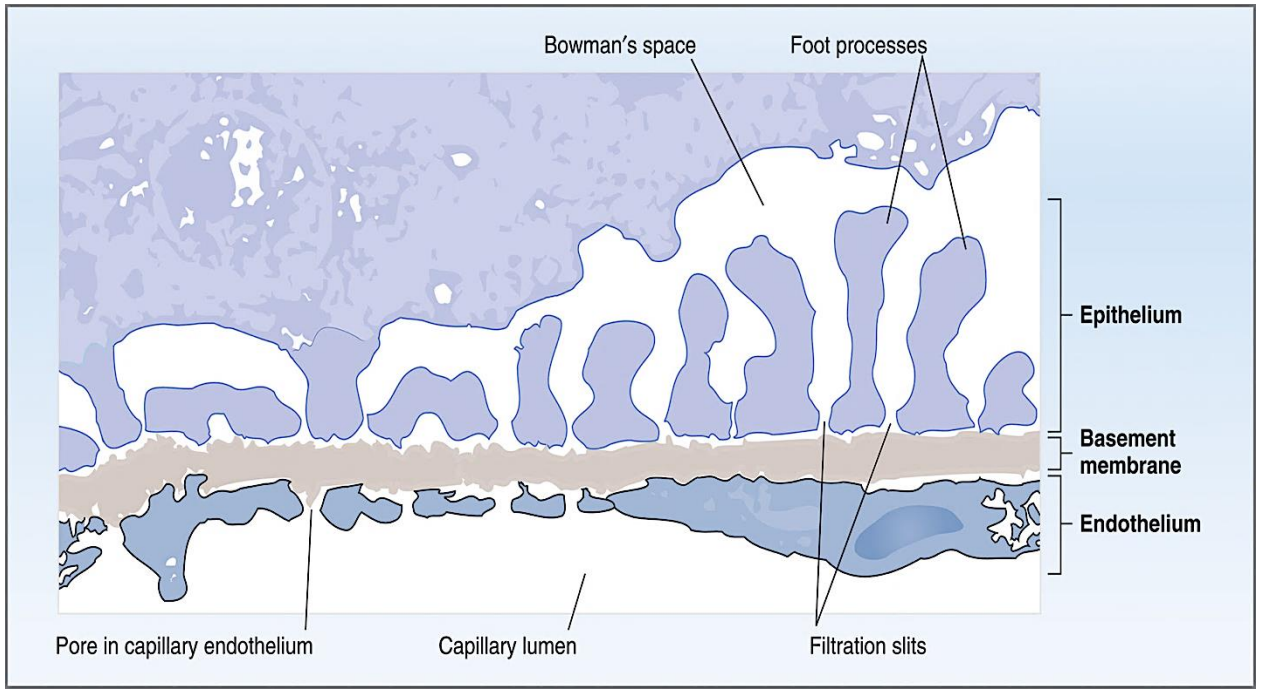
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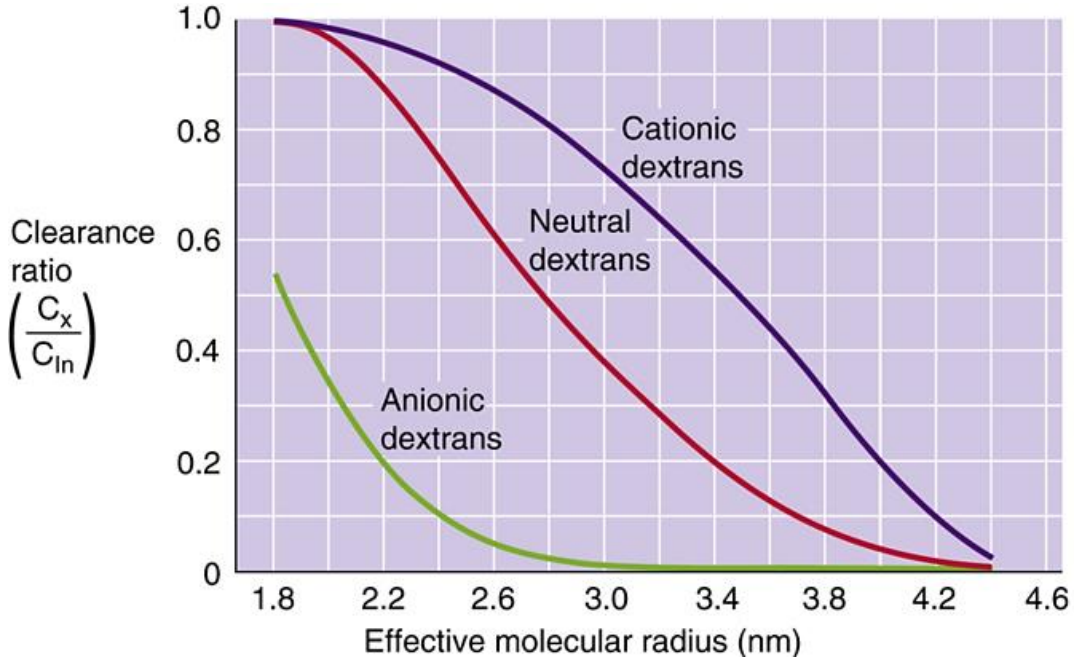
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Glomerular Filtration Barrier: Size & Charge Selectivity

- **The Tri-Layer Sieve:** Ultrafiltration requires crossing fenestrated capillary endothelium, the acellular basement membrane, and podocyte filtration slits in series.
- **Size Selectivity:**
  - Molecules with molecular weight $< 5{,}500$ Da (e.g., water, Na^+, glucose, inulin) filter freely (filterability = 1.0).
  - Molecules $> 69{,}000$ Da are effectively excluded under normal conditions.
- **Charge Selectivity:** Fixed negative charges across the GBM and slit diaphragms repel polyanions (e.g., plasma albumin, MW 69 kDa, effective radius ~ 3.6 nm, filterability ~ 0.005).
- **Clinical Correlation:**
  - *Nephrotic Syndrome:* Selective loss of GBM negative charges leads to heavy proteinuria (albuminuria), hypoalbuminemia, and generalized edema.
  - *Nephritic Syndrome:* Inflammatory architectural disruption of the barrier, permitting passage of both red blood cells (hematuria/RBC casts) and protein.

Glomerular Capillary Wall and Size/Charge Selectivity





Driving Forces of GFR: Starling Forces & Net Filtration Pressure

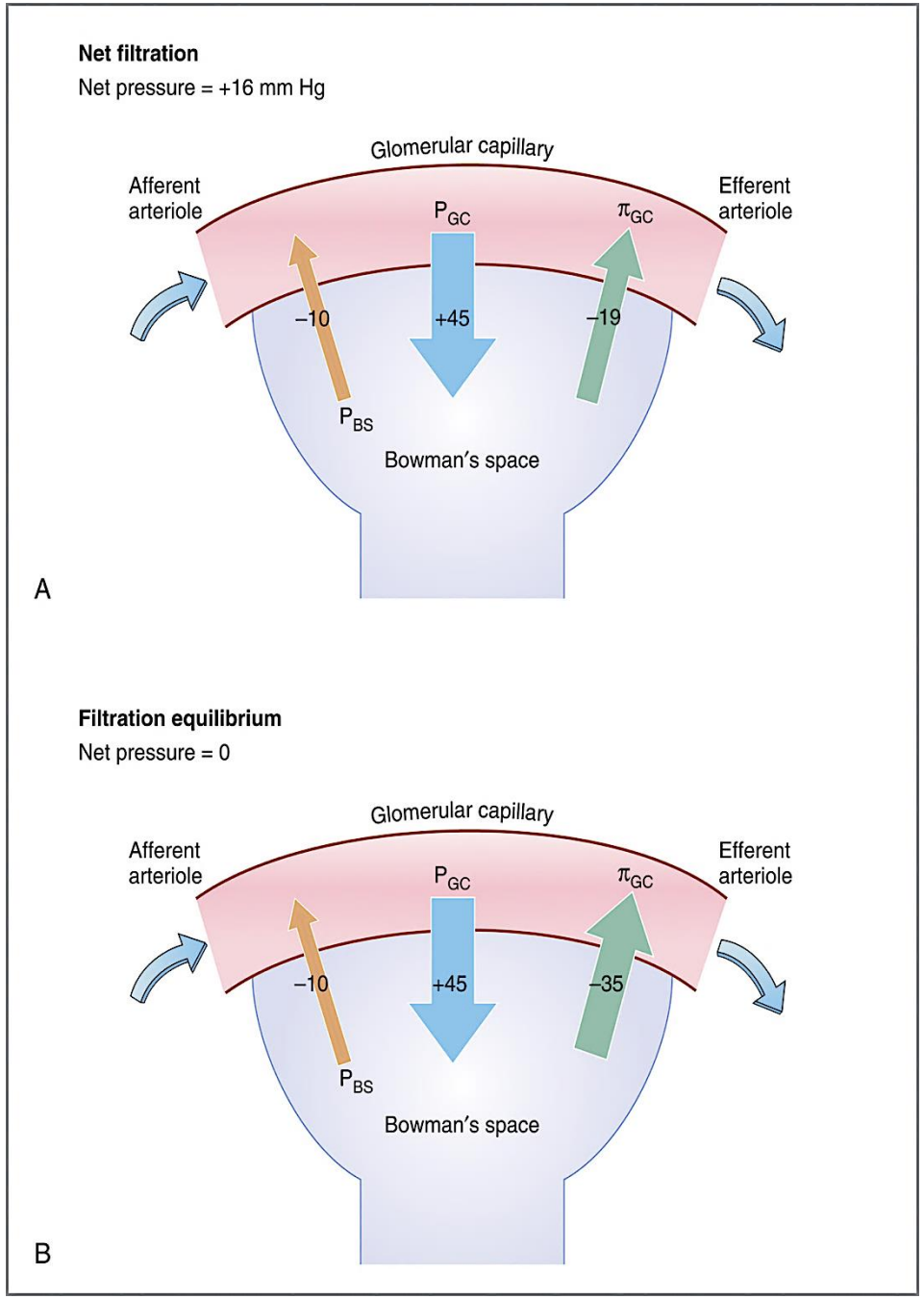
$GFR = K_f[(P_{GC} - P_{BS}) - \pi_{GC}]$, where:

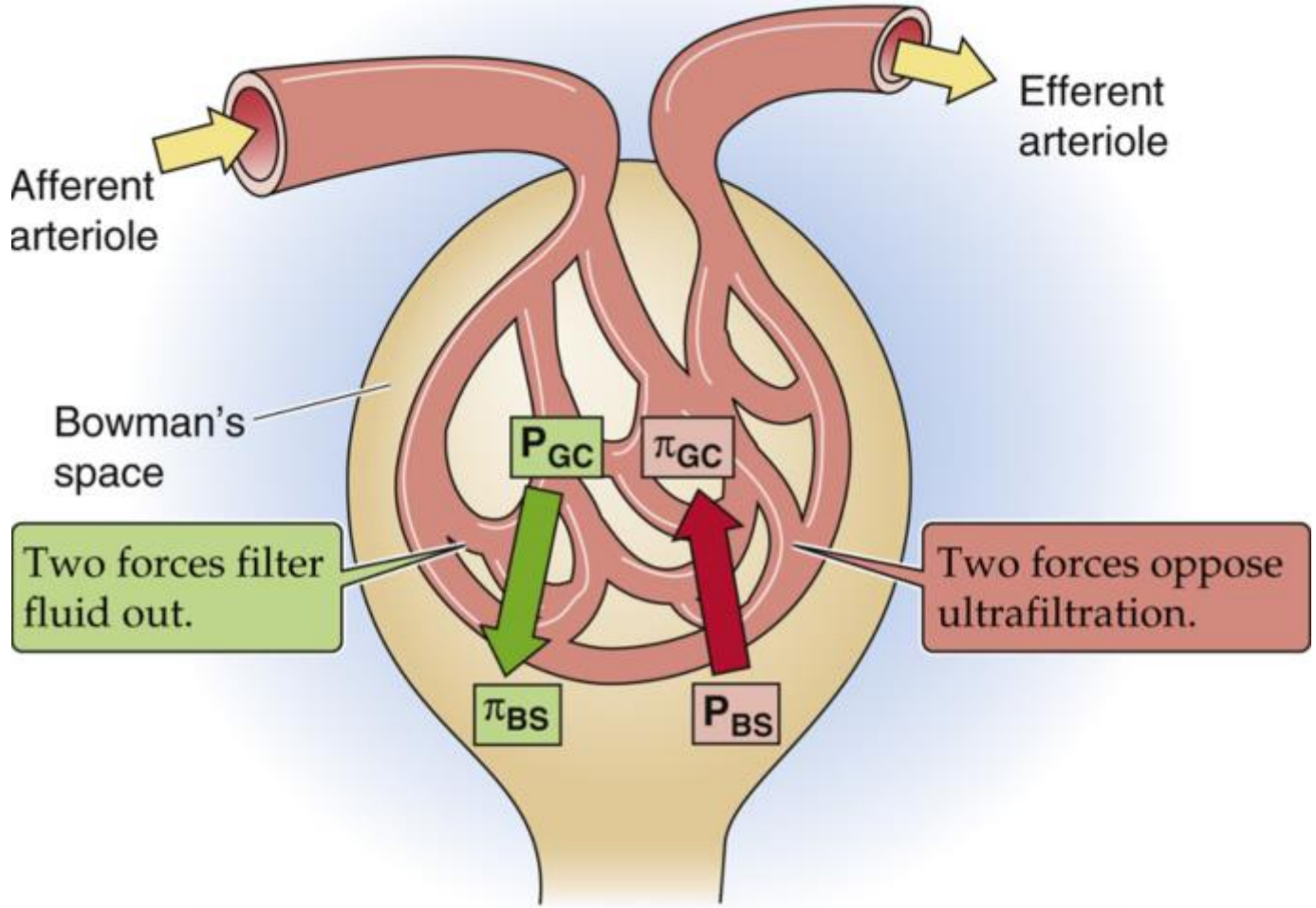
- GFR = Glomerular filtration rate (mL/min)
- K_f = Hydraulic conductance (mL/min/mm Hg)
- **K_f, the filtration coefficient,** is the water permeability or hydraulic conductance of the glomerular capillary wall.
- Two factors that contribute to K_f are the water permeability per unit of surface area and the total surface area.
- K_f for glomerular capillaries is more than 100-fold that for systemic capillaries.
- The consequence of this extremely high K_f is that much more fluid is filtered from glomerular capillaries than from other capillaries (i.e., GFR is 180 L/day).

Driving Forces of GFR: Starling Forces & Net Filtration Pressure (Continued)

$GFR = K_f[(P_{GC} - P_{BS}) - \pi_{GC}]$, where:

- **P_{GC}, hydrostatic pressure in glomerular capillaries,** is a force favoring filtration.
  - Compared with systemic capillaries, P_{GC} is relatively high (45 mm Hg).
- **P_{BS}, hydrostatic pressure in Bowman's space,** is a force opposing filtration.
  - The origin of this pressure (10 mm Hg) is the fluid present in the lumen of the nephron.
- **π_{GC}, oncotic pressure in glomerular capillaries,** is a force opposing filtration
  - **π_{GC} determined by the protein concentration of glomerular capillary blood.**
  - π_{GC} *does not remain constant* along the capillary length, it progressively increases as fluid is filtered out of the capillary.
  - Eventually, π_{GC} increases to the point where net ultrafiltration pressure becomes zero and glomerular filtration stops (called filtration equilibrium).



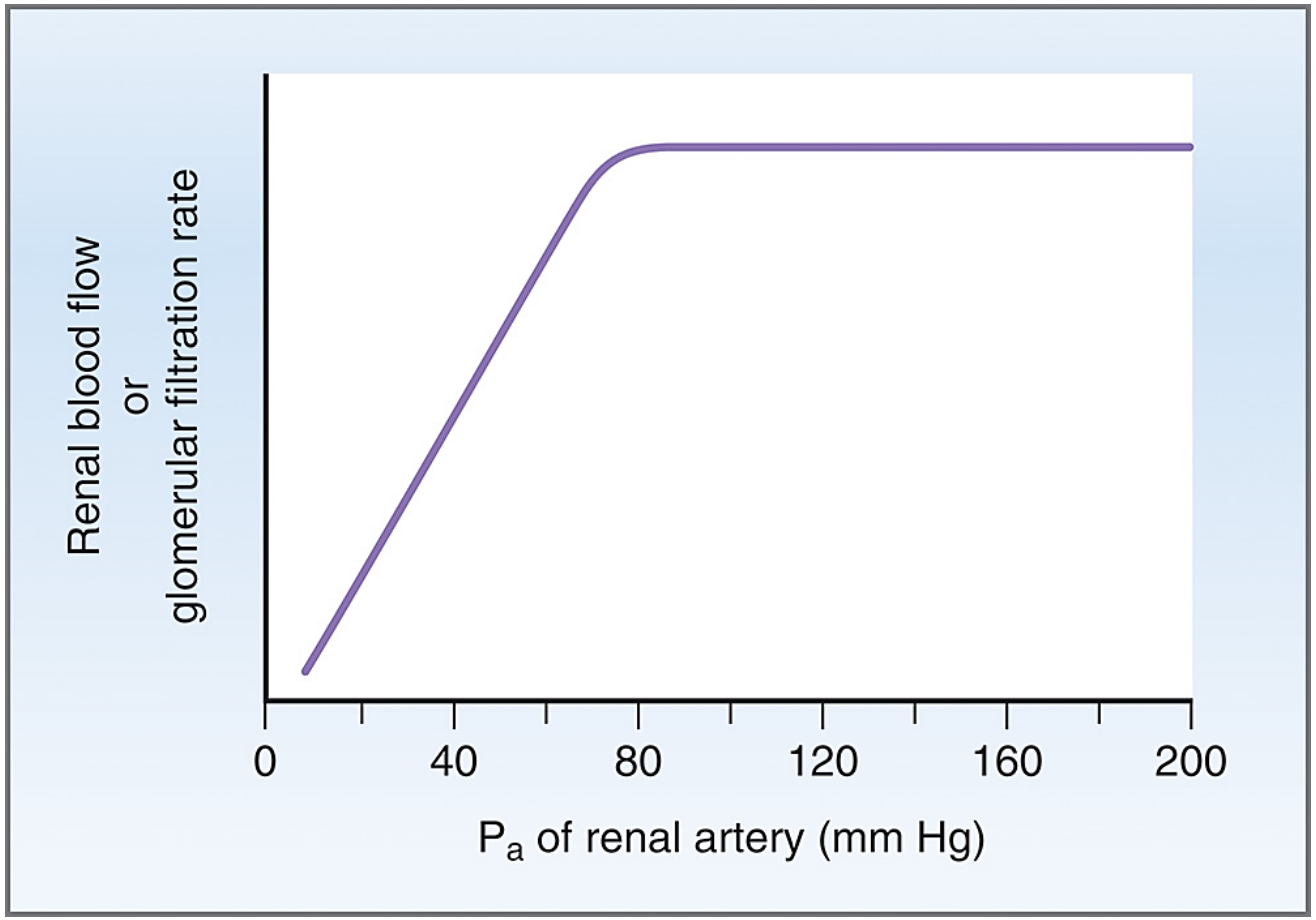


P_{GC} = Glomerular capillary hydrostatic pressure
π_{BS} = Bowman's space oncotic pressure
P_{BS} = Bowman's space hydrostatic pressure
π_{GC} = Glomerular capillary oncotic pressure

Intrinsic GFR Autoregulation: The Myogenic Response

- **The Purpose of Autoregulation:** Keeps renal blood flow (RBF) and GFR relatively constant across systemic mean arterial pressures between 80 and 180 mmHg.
  - Protects delicate glomerular capillaries from hypertensive injury and prevents excessive salt and water loss.
- **The Myogenic Mechanism:** An intrinsic vascular response initiated directly by vascular smooth muscle in afferent arterioles.
- **Response to Increased Blood Pressure:**
  - ↑ MAP → transmural stretch of afferent arteriole
  - → opening of stretch–activated cation channels → depolarization
  - → vasoconstriction of afferent arteriole ($\uparrow R_a$)
  - → blunts blood flow into glomerulus → P_G maintained at 60 mmHg → GFR stabilized at 125 mL/min.
- **Response to Decreased Blood Pressure:** Reduced vascular stretch causes passive afferent vasodilation, preserving capillary perfusion and sustaining GFR.

Autoregulation

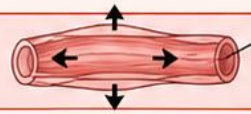


Myogenic Autoregulation of Renal Blood Flow and GFR

Response to an ACUTE INCREASE in Mean Arterial Pressure (MAP)

Acute ↑ MAP

↑ Transmural pressure and stretch of afferent arteriolar smooth muscle



Stretched afferent arteriole (smooth muscle)

Stretch-activated cation channels open → depolarization

↑ Ca^{2+} influx into smooth muscle

Afferent arteriolar **vasoconstriction** → ↑ afferent arteriolar resistance (R_a)

Shields the glomerulus: P_{GC} (glomerular capillary hydrostatic pressure) remains ~55–60 mmHg

RBF and GFR remain relatively stable

Response to an ACUTE DECREASE in Mean Arterial Pressure (MAP)

Acute ↓ MAP

↓ Stretch of afferent arteriolar smooth muscle



Less stretched afferent arteriole (smooth muscle)

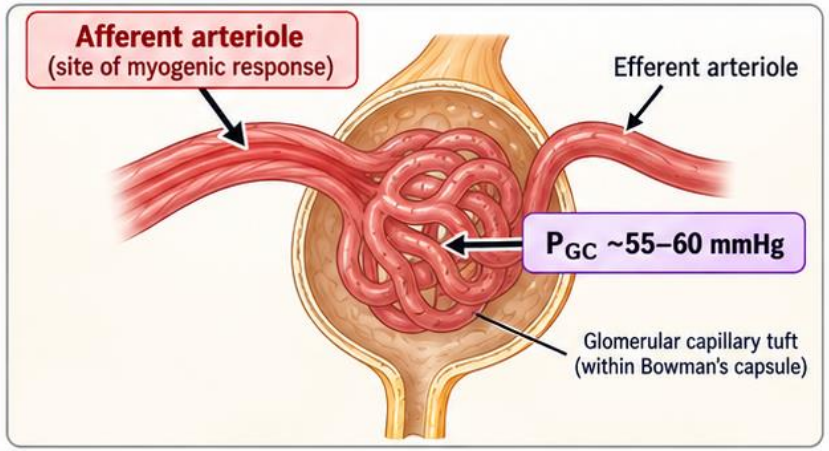
Less activation of stretch-sensitive channels → less depolarization

↓ Ca^{2+} influx

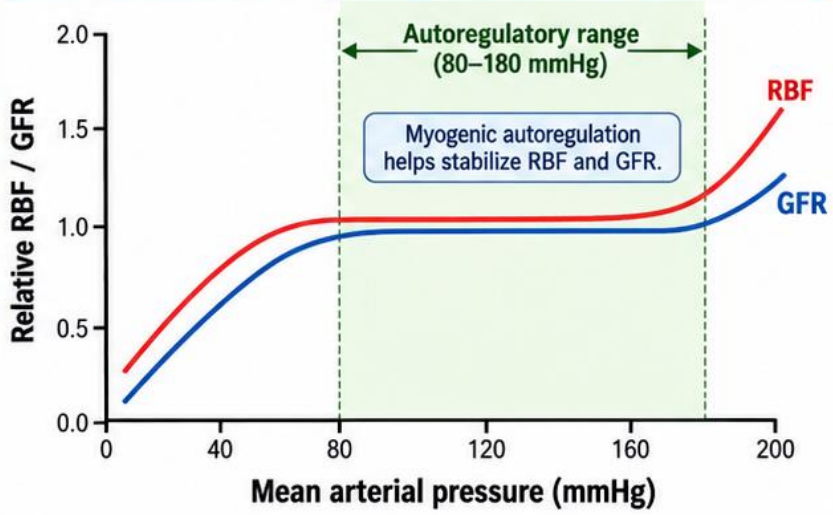
Afferent arteriolar **vasodilation** → ↓ afferent arteriolar resistance (R_a)

Helps maintain P_{GC} (glomerular capillary hydrostatic pressure) near ~55–60 mmHg

RBF and GFR remain relatively stable



Renal Autoregulation of RBF and GFR

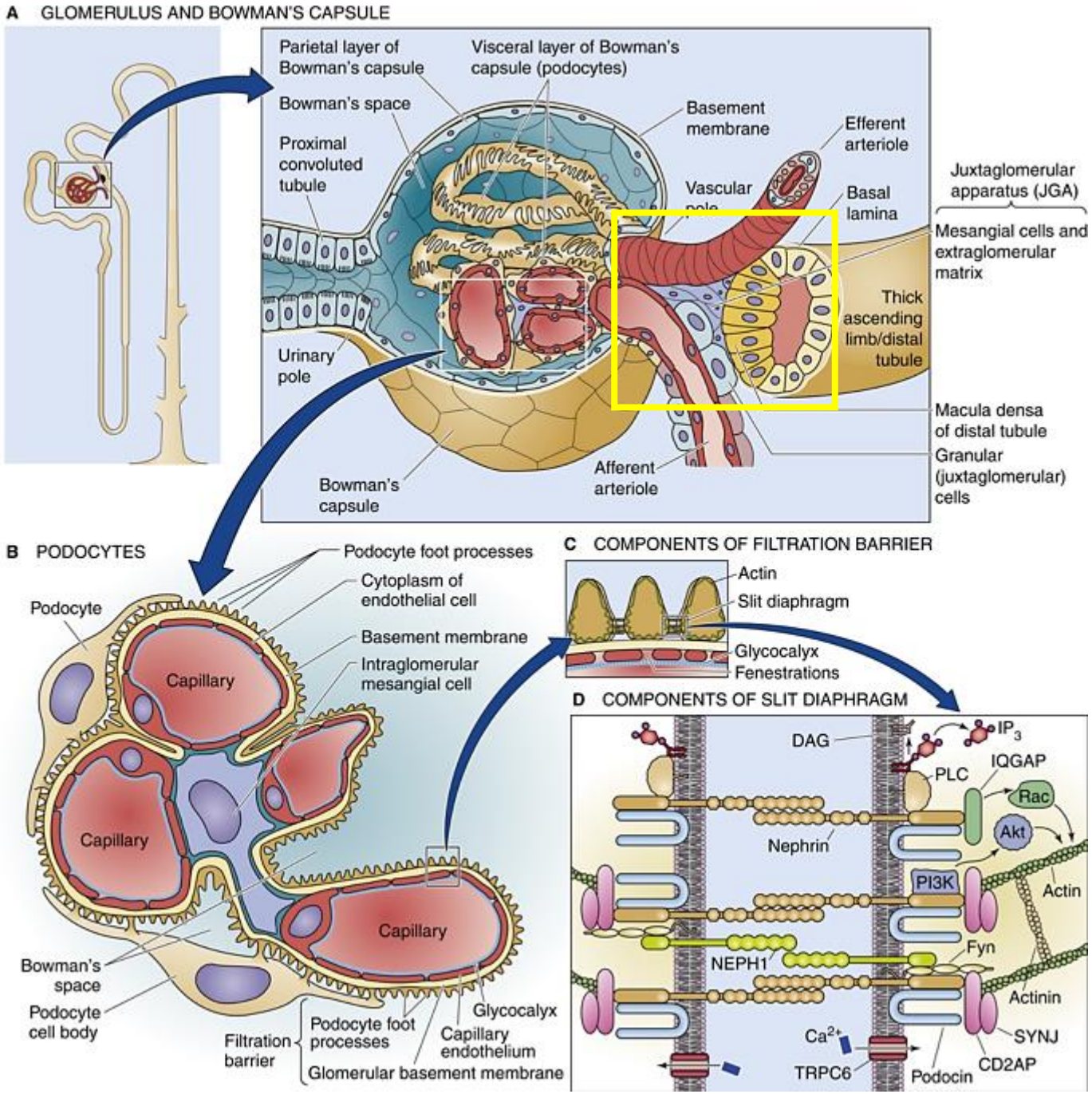


Key Points

- **Afferent arteriole = primary resistance vessel** responding to acute pressure changes.
- **Stretch → Ca^{2+}-dependent smooth muscle contraction.**
- This myogenic response **buffers transmission** of systemic pressure to the glomerular capillaries.

GFR Autoregulation: Tubuloglomerular Feedback

- **Tubuloglomerular Feedback (TGF) Defined:** An intrinsic negative feedback loop wherein solute delivery to the distal nephron regulates upstream glomerular filtration rate.
- **Sensor Mechanism: Macula densa** cells detect luminal fluid flow rate via Na^+–K^+–$2Cl^-$ (NKCC2) uptake.
- **Sequence During Elevated Perfusion / High GFR:**
  - Acute ↑ GFR →↑ flow of tubular fluid past macula densa.
  - ↑ NaCl delivery and uptake by macula densa cells.
  - Release of paracrine signaling molecules: **ATP and Adenosine**.
  - Binding to A_1 receptors on afferent arteriolar smooth muscle → **vasoconstriction**.
  - Downstream P_G decreases to 60 mmHg → GFR returns to baseline $(125$ mL/min$)$.
- **Sequence During Reduced Perfusion / Low GFR:**
  - ↓ NaCl delivery to macula densa → release of prostaglandins (PGE_2) and NO → afferent arteriole dilates and granular cells secrete **Renin**.


A GLOMERULUS AND BOWMAN'S CAPSULE
Parietal layer of Bowman's capsule
Visceral layer of Bowman's capsule (podocytes)
Bowman's space
Basement membrane
Efferent arteriole
Proximal convoluted tubule
Vascular pole
Basal lamina
Juxtaglomerular apparatus (JGA)
Mesangial cells and extraglomerular matrix
Thick ascending limb/distal tubule
Urinary pole
Macula densa of distal tubule
Granular (juxtaglomerular) cells
Bowman's capsule
Afferent arteriole
B PODOCYTES
Podocyte foot processes
Cytoplasm of endothelial cell
Podocyte
Basement membrane
Intraglomerular mesangial cell
Capillary
Capillary
Capillary
Bowman's space
Podocyte cell body
Glycocalyx
Filtration barrier
Podocyte foot processes
Capillary endothelium
Glomerular basement membrane
C COMPONENTS OF FILTRATION BARRIER
Actin
Slit diaphragm
Glycocalyx
Fenestrations
D COMPONENTS OF SLIT DIAPHRAGM
DAG
IP3
PLC
IQGAP
Rac
Akt
Nephrin
PI3K
Actin
Fyn
NEPH1
Actinin
Ca2+
SYNJ
TRPC6
Podocin
CD2AP

Interim Summary 2

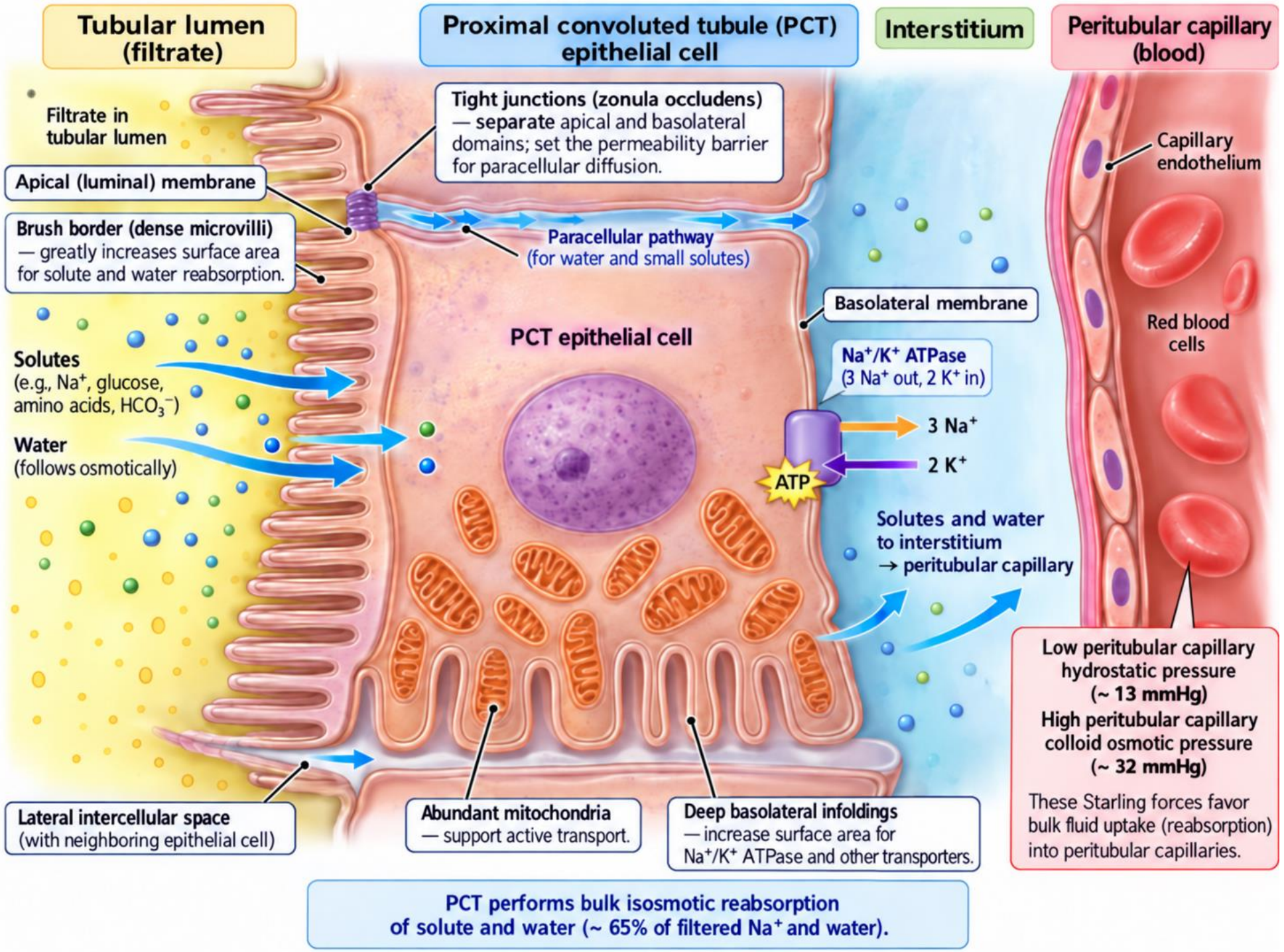
- **The Tri-Layer Sieve:** Fenestrated endothelium stops cells, while the negatively charged glomerular basement membrane (GBM) and podocyte slit diaphragms electrostatically repel polyanionic proteins like albumin.
- **Starling Forces & Filtration Equilibrium:** Net ultrafiltration is driven by high capillary hydrostatic pressure (P_{GC}) opposing Bowman's hydrostatic pressure (P_{BS}) and rising capillary oncotic pressure (π_{GC}). In humans, π_{GC} climbs until net pressure approaches zero (filtration equilibrium) near the efferent end.
- **Intrinsic Autoregulation ($80-180$ mmHg):**
  - *Myogenic Reflex:* Afferent vascular smooth muscle contracts in response to transmural stretch to shield downstream capillaries from hypertensive spikes.
  - *Tubuloglomerular Feedback (TGF):* Macula densa cells sense increased tubular flow/NaCl delivery via NKCC2 and release adenosine/ATP to constrict the afferent arteriole, stabilizing GFR.

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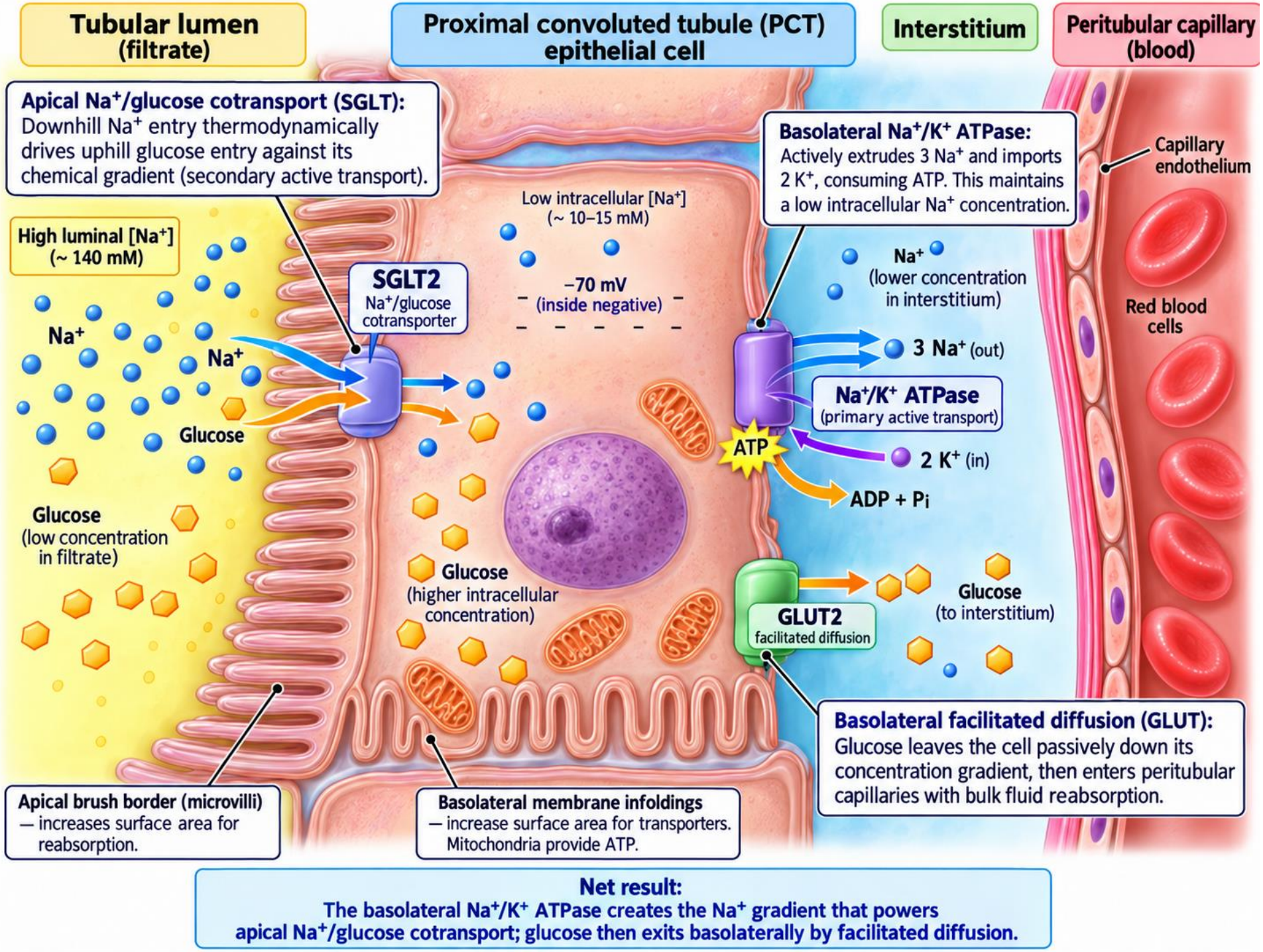
Proximal Convoluted Tubule (PCT)

- **Epithelial Polarity Enables Directional Transport:** Distinct apical and basolateral protein distributions prevent back-leak and drive unidirectional reabsorption.
- **Two Transport Pathways:**
  - *Transcellular:* Crosses apical membrane, cytoplasm, and basolateral membrane.
  - *Paracellular:* Crosses tight junctions through intercellular spaces.
- **The Nephron's Bulk Reabsorber:** Unregulated retrieval of roughly two-thirds ($\sim 65–67\%$) of filtered Na^+ and water, alongside virtually 100% of filtered glucose and amino acids.


Tubular lumen (filtrate)
Proximal convoluted tubule (PCT) epithelial cell
Interstitium
Peritubular capillary (blood)
Filtrate in tubular lumen
Apical (luminal) membrane
Brush border (dense microvilli)
— greatly increases surface area for solute and water reabsorption.
Tight junctions (zonula occludens)
— separate apical and basolateral domains; set the permeability barrier for paracellular diffusion.
Paracellular pathway
(for water and small solutes)
Basolateral membrane
PCT epithelial cell
Solutes
(e.g., Na+, glucose, amino acids, HCO3−)
Water
(follows osmotically)
Na+/K+ ATPase
(3 Na+ out, 2 K+ in)
3 Na+
2 K+
ATP
Solutes and water to interstitium → peritubular capillary
Capillary endothelium
Red blood cells
Low peritubular capillary hydrostatic pressure (~ 13 mmHg)
High peritubular capillary colloid osmotic pressure (~ 32 mmHg)
These Starling forces favor bulk fluid uptake (reabsorption) into peritubular capillaries.
Lateral intercellular space
(with neighboring epithelial cell)
Abundant mitochondria
— support active transport.
Deep basolateral infoldings
— increase surface area for Na+/K+ ATPase and other transporters.
PCT performs bulk isosmotic reabsorption of solute and water (~ 65% of filtered Na+ and water).

PCT: Reabsorption Mechanics

- **The Basolateral Pump is the Primary Engine:** The Na^+/K^+-ATPase maintains low intracellular [Na^+] to generate the chemical energy driving tubular reabsorption.
- **Secondary Active Transport Example:** Luminal glucose reabsorption requires no direct ATP hydrolysis at the apical membrane; it runs entirely on the Na^+ gradient established by the basolateral pump.
- **Isosmotic Fluid Reabsorption:** High water permeability (via constitutive AQP1 and paracellular channels) ensures water follows solute equimolarly, keeping proximal tubular fluid isosmotic (~ 300 mOsm/L).


Tubular lumen
(filtrate)
Proximal convoluted tubule (PCT) epithelial cell
Interstitium
Peritubular capillary
(blood)
Apical Na+/glucose cotransport (SGLT):
Downhill Na+ entry thermodynamically drives uphill glucose entry against its chemical gradient (secondary active transport).
Basolateral Na+/K+ ATPase:
Actively extrudes 3 Na+ and imports 2 K+, consuming ATP. This maintains a low intracellular Na+ concentration.
Capillary endothelium
Red blood cells
High luminal [Na+]
(~ 140 mM)
Low intracellular [Na+]
(~ 10–15 mM)
SGLT2
Na+/glucose cotransporter
−70 mV
(inside negative)
Na+
(lower concentration in interstitium)
Na+
Na+
Glucose
3 Na+ (out)
Na+/K+ ATPase
(primary active transport)
ATP
2 K+ (in)
ADP + Pi
Glucose
(low concentration in filtrate)
Glucose
(higher intracellular concentration)
GLUT2
facilitated diffusion
Glucose
(to interstitium)
Basolateral facilitated diffusion (GLUT):
Glucose leaves the cell passively down its concentration gradient, then enters peritubular capillaries with bulk fluid reabsorption.
Apical brush border (microvilli)
— increases surface area for reabsorption.
Basolateral membrane infoldings
— increase surface area for transporters. Mitochondria provide ATP.
Net result:
The basolateral Na+/K+ ATPase creates the Na+ gradient that powers apical Na+/glucose cotransport; glucose then exits basolaterally by facilitated diffusion.

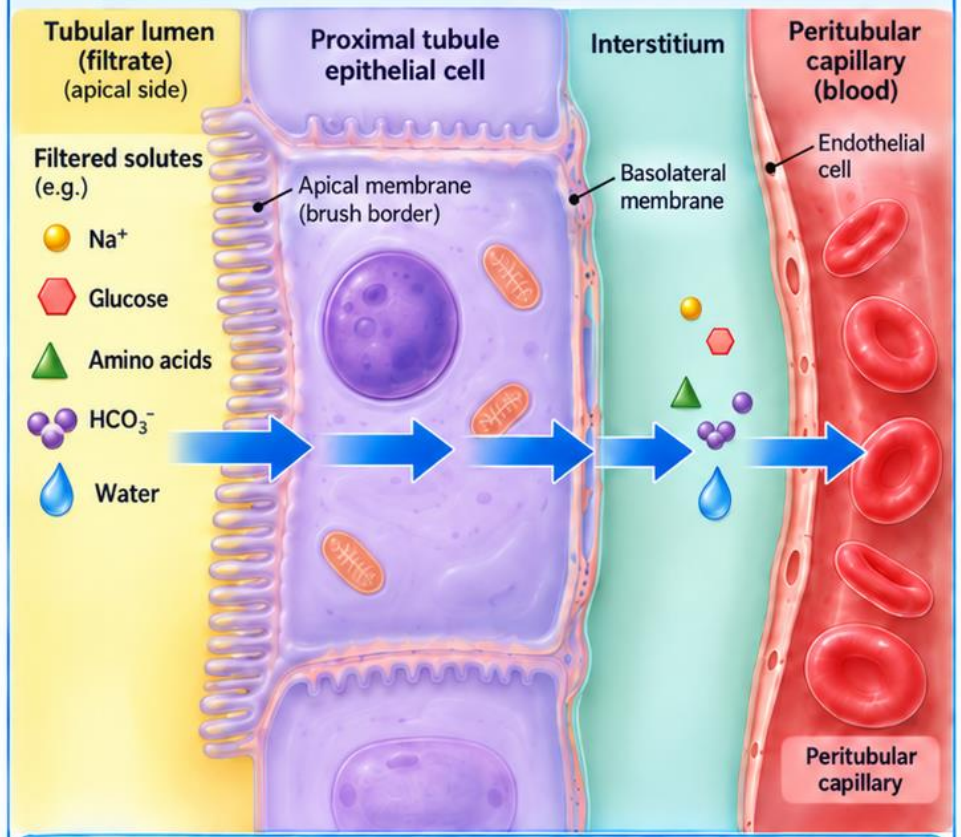
Tubular Secretion Mechanics & Net Excretion Balance

- **Secretion Defined:** Direct active extrusion of substances from peritubular capillaries into the tubular lumen, bypassing the glomerular filter.
- **Primary Physiological Role:** Eliminates protein-bound toxins, foreign metabolites, and xenobiotics that cannot be filtered at the glomerulus.
- **Clinical Archetypes:**
  - *Penicillin Clearance:* Rapidly secreted via proximal OAT carriers; probenecid competitively blocks OAT to prolong antibiotic half-life.
  - *Creatinine:* Freely filtered and slightly secreted ($\sim 10–20\%$), making creatinine clearance slightly higher than actual GFR.
- **Impact on the Mass Balance Equation:**
  - Net Reabsorption: Excretion $<$ Filtration ($E = F - R$)
  - Net Secretion: Excretion $>$ Filtration ($E = F + S$)

Reabsorption

Net movement: lumen to blood

Reabsorption returns filtered substances from tubular fluid to the blood.



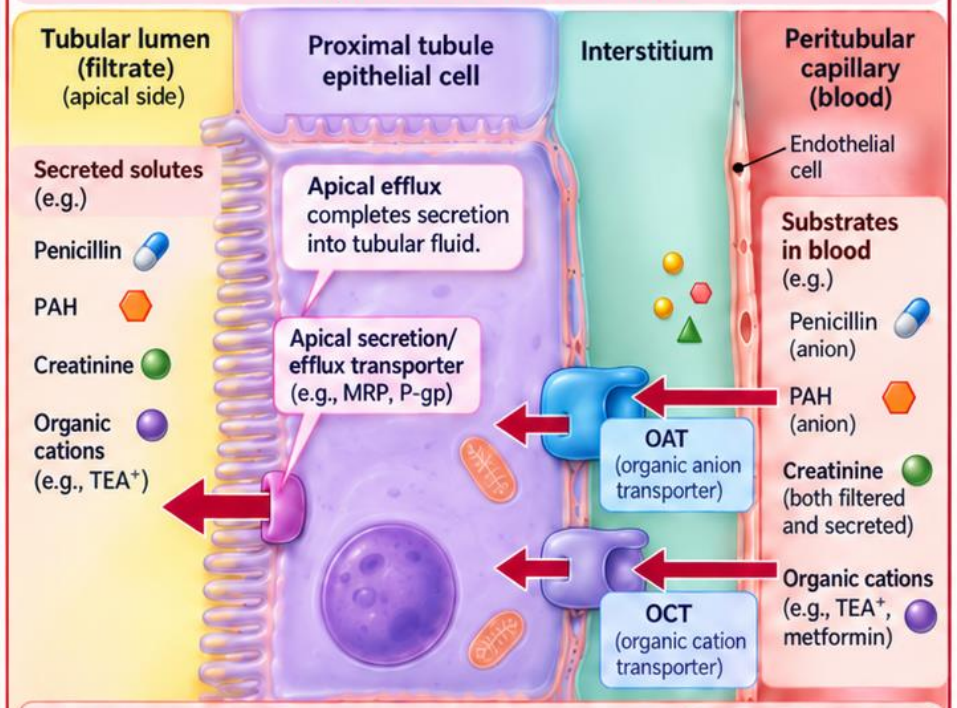
Excretion = Filtration − Reabsorption + Secretion

If reabsorption predominates, E < F

Reabsorption reduces urinary excretion relative to the filtered load.

Tubular secretion

Secretion vector: blood → interstitium → basolateral uptake → apical efflux → tubular fluid



- Basolateral OAT/OCT systems move selected organic anions/cations from peritubular blood into the tubular cell.
- Apical efflux completes secretion into tubular fluid.
- Secretion is especially important for compounds that are substantially protein-bound in the circulation and therefore not freely filtered. Only the free fraction is transported.
- Creatinine is filtered and also slightly secreted.
- Penicillin secretion can be reduced by probenecid.

Secretion permits efficient elimination of drugs that are poorly filtered because **much** of the circulating solute is protein-bound. Only the free fraction is secreted; bound drug dissociates to replenish the free pool.

Excretion = Filtration − Reabsorption + Secretion

When secretion adds solute to the lumen, excretion can exceed the filtered load: **E > F**

Secretion permits elimination of protein-bound or poorly filtered solutes/drugs.

Reabsorption moves solute from lumen to blood; secretion moves solute from blood to lumen.
Secretion can increase urinary excretion: above the filtered load.

Interim Summary 3

- **Basolateral Na^+/K^+-ATPase is the Engine:** Primary active extrusion of 3 Na^+ for 2 K^+ maintains low intracellular $[Na^+]$ and negative membrane voltage, fueling all secondary active transport across the apical membrane.
- **Unregulated Bulk Reabsorption:** The proximal tubule is an isosmotic workhorse, reabsorbing $\sim 65–67\%$ of filtered Na^+ and water, and virtually 100% of filtered glucose (via SGLT2/1) and amino acids.
- **Secretion as an Active Elimination Route:** Peritubular transporters (basolateral OAT/OCT and apical efflux pumps) actively pump protein-bound drugs (e.g., penicillin) and end-products (creatinine) into the lumen, driving excretion above filtration ($E > F$).

Learning Objectives

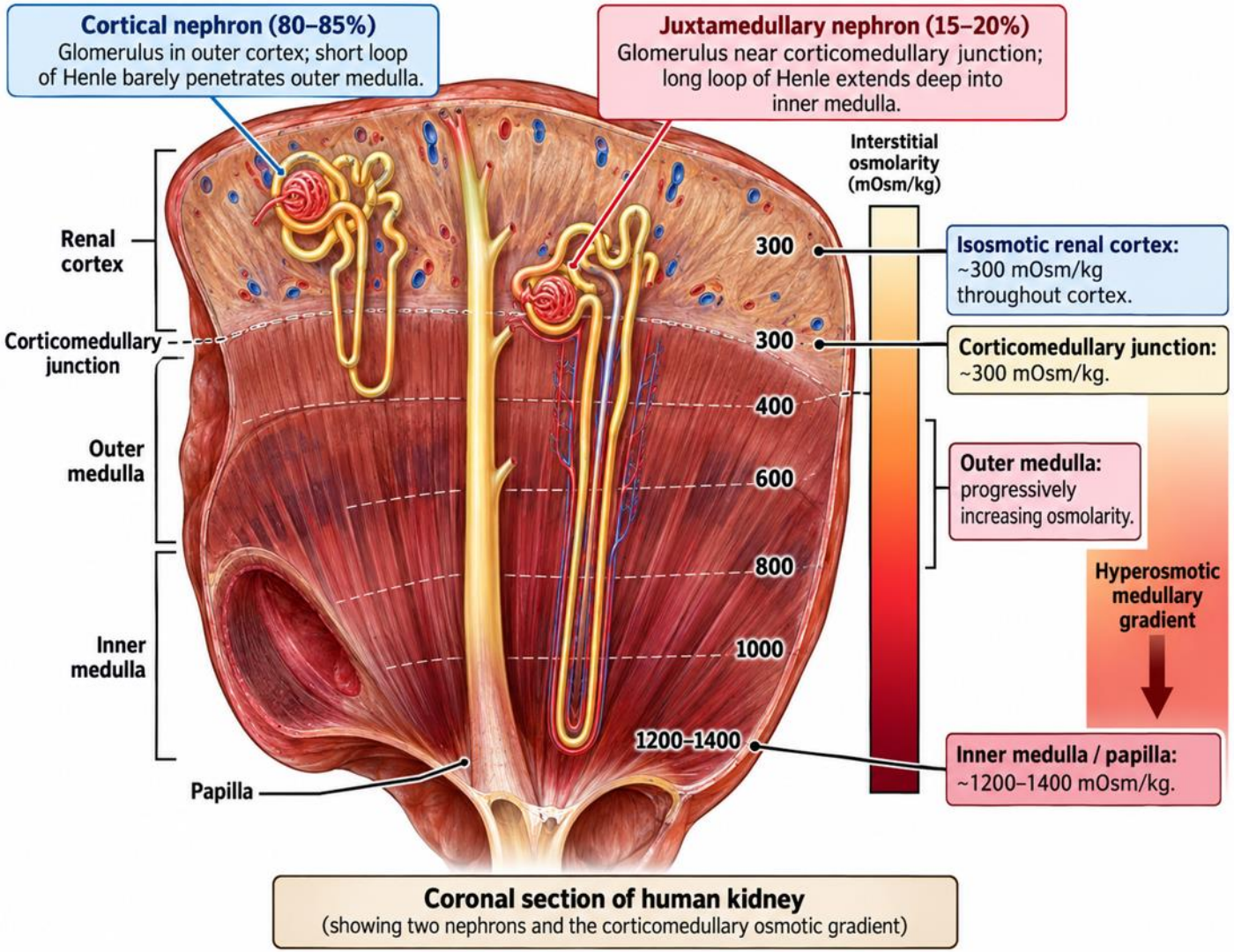
1. Correlate the microanatomical structure of the nephron and its portal microcirculation with renal plasma flow and the fundamental mass balance equation.
2. Evaluate how the glomerular filtration barrier, Starling forces, and intrinsic autoregulatory mechanisms determine net filtration pressure and glomerular filtration rate.
3. Characterize the general epithelial transport mechanisms of proximal tubular reabsorption and peritubular secretion, highlighting their opposing roles in determining net excretion.
4. Describe how differing transport properties of the descending and ascending limbs of the Loop of Henle establish the hyperosmotic medullary gradient.
5. Explain how final urinary excretion and concentration are modulated in the collecting duct by the presence or absence of ADH and aldosterone.

The Corticomedullary Osmotic Gradient & Nephron Anatomy

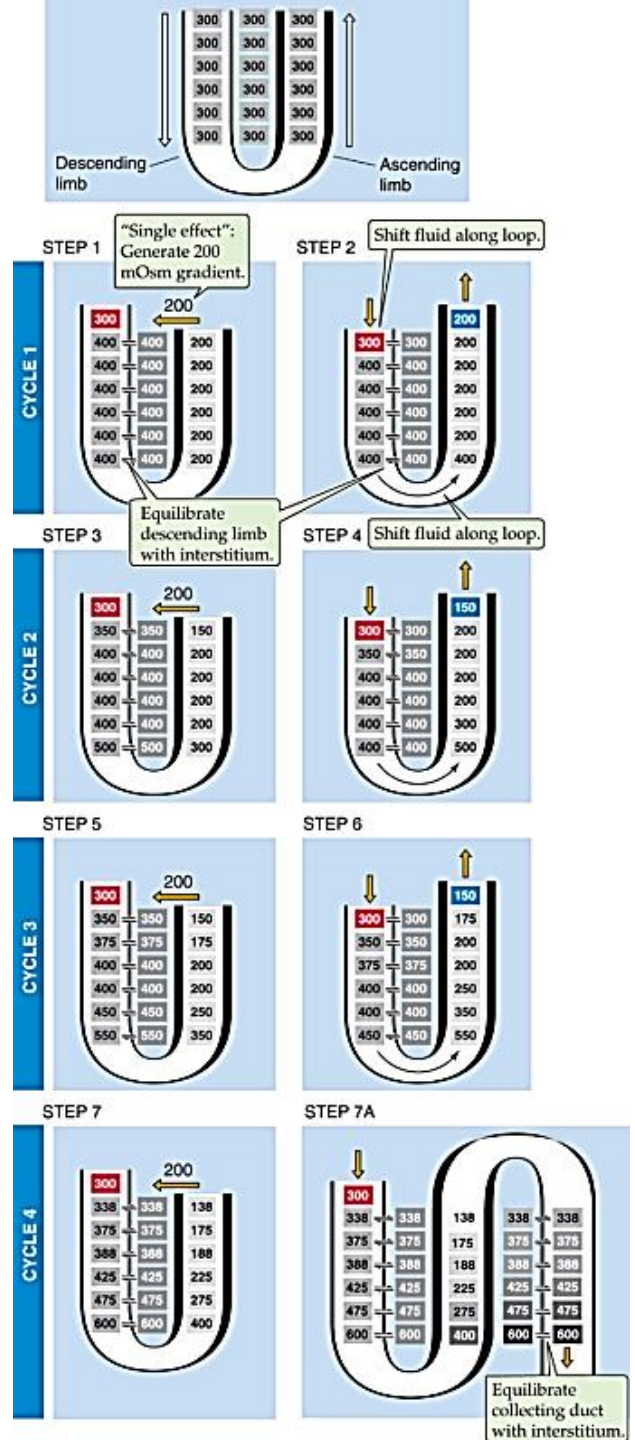
- **The Medullary Gradient Defined:** An osmotic stratification of the renal interstitium that becomes progressively hypertonic toward the renal pelvis.
- **Architects of the Gradient:** Juxtamedullary nephrons generate this gradient because their long loops of Henle extend deep into the inner medulla.
- **Division of Labor:**
  - *The Loop of Henle:* **Produces** and multiplies the medullary concentration gradient.
  - *The Collecting Duct:* **Uses** the osmotic gradient to concentrate final urine under the control of ADH.
- **Physiological Range:** Enables the kidney to excrete urine across a wide concentration spectrum, from 50 mOsm/kg (extreme dilution) to 1200 mOsm/kg (maximal concentration).

Differing Transport Properties: Descending vs. Ascending Limbs of Loop of Henle

- **Asymmetric Permeability Properties:**
  - *Descending Limb:* Highly permeable to water, virtually impermeable to NaCl and other solutes; water leaves via osmosis, concentrating tubular fluid toward the hairpin turn.
  - *Ascending Limb:* Impermeable to water, actively reabsorbs Na^+, K^+, and Cl^-; solute leaves without water, diluting tubular fluid and salting the interstitium.
- **The "Single Effect":** Active solute extrusion by the thick ascending limb creates a horizontal gradient of up to 200 mOsm/kg between the tubular lumen and the adjacent interstitium at any given horizontal level.
- **Countercurrent Multiplication:** Countercurrent fluid flow (down the descending limb, up the ascending limb) continually multiplies that 200 mOsm/kg horizontal gradient into a large 1100 mOsm/kg longitudinal gradient from cortex to papilla (300 → 1200 mOsm/kg).
- **Net Result:** Medullary interstitial tissue is kept hypertonic, while the fluid exiting the loop to the distal tubule is made hypoosmotic (50–100 mOsm/kg).

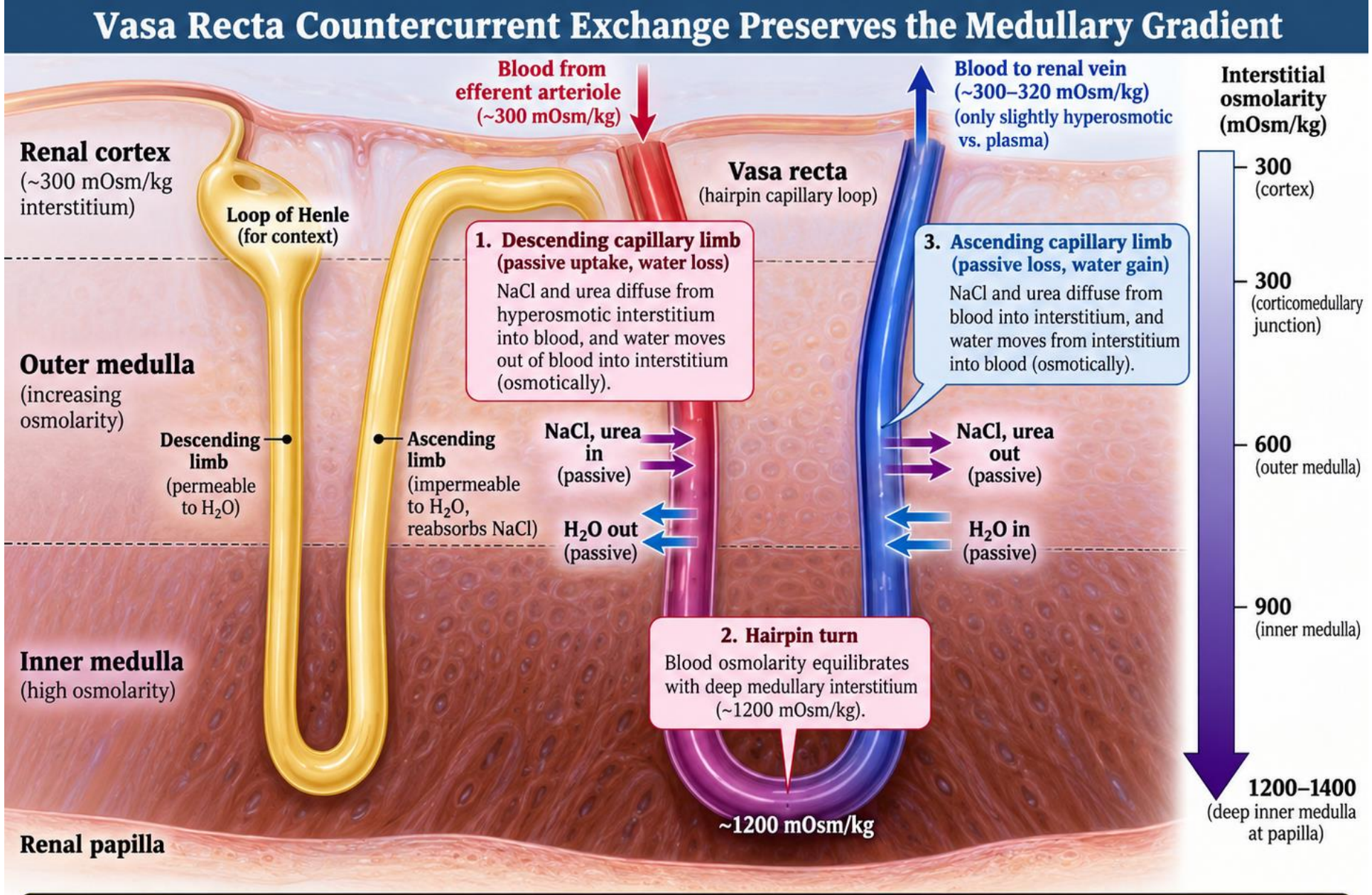


Coronal section of human kidney
(showing two nephrons and the corticomedullary osmotic gradient)



Preserving the Gradient: The Vasa Recta as Countercurrent Exchangers

- **The Challenge of Medullary Perfusion:** Medullary tissue requires blood flow for oxygen and nutrient delivery, but excessive or rapid blood flow would wash away the hypertonic solute gradient.
- **Countercurrent Exchange (Passive Mechanism):**
  - The vasa recta do not create the medullary gradient; they **passively preserve** it.
  - Highly permeable hairpin capillary loops parallel the loops of Henle.
- **Two-Way Passive Transit:**
  - *Descending into medulla:* Blood absorbs interstitial solutes and loses water, equilibrating with the concentrated tissue.
  - *Ascending toward cortex:* Blood reabsorbs water and releases solutes back into the medullary interstitium.
- **Low Blood Flow Rate:** Sluggish blood flow through the renal medulla minimizes solute removal, ensuring the hyperosmotic environment remains intact.



Summary: The vasa recta function as **countercurrent exchangers:** they preserve, rather than generate, the corticomedullary osmotic gradient while returning reabsorbed water to the circulation.

Interim Summary 4

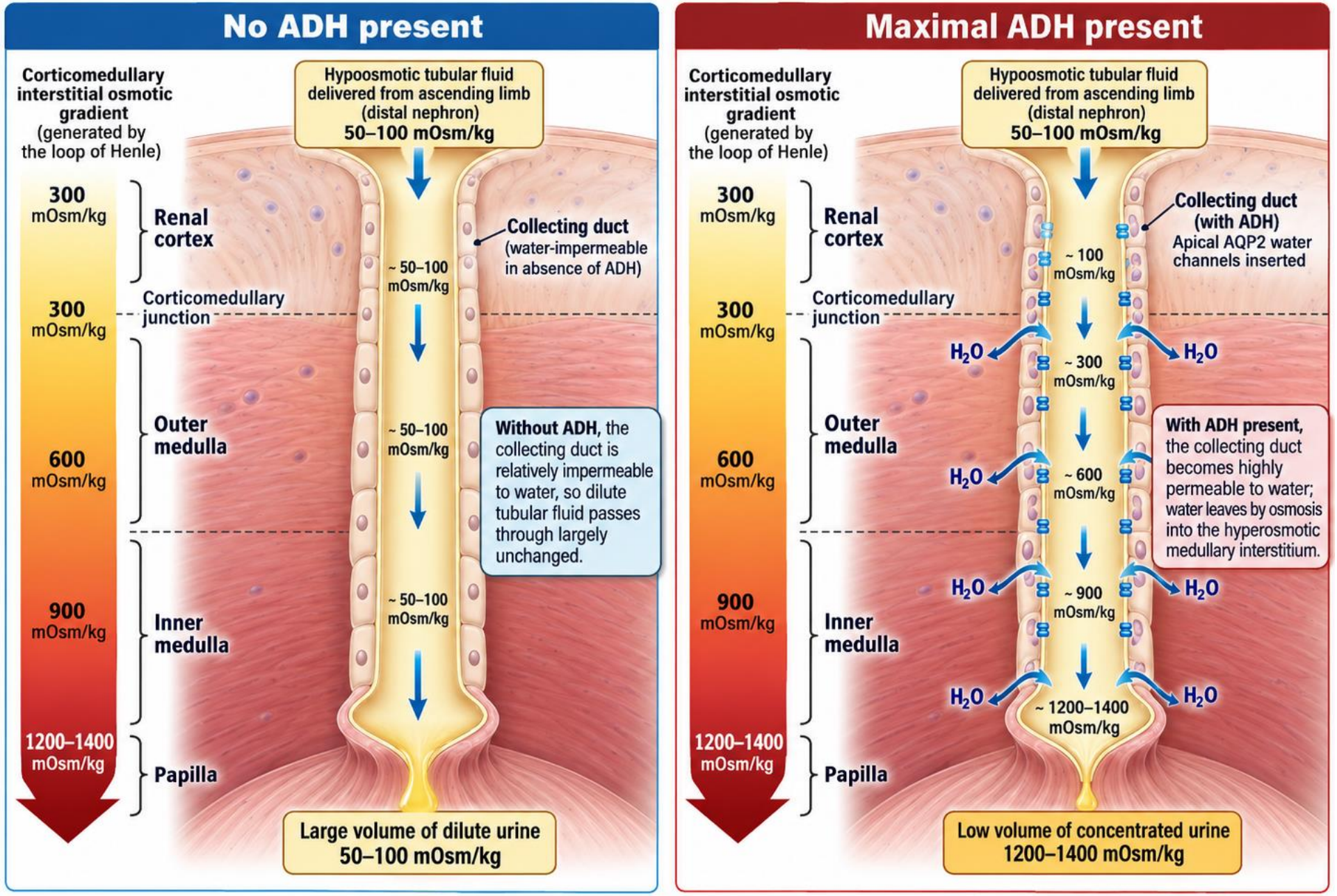
- **Architects vs. Preservers:** Long-looped juxtamedullary nephrons **generate** the hyperosmotic corticomedullary gradient ($300 \rightarrow 1200–1400$ mOsm/kg), while the hairpin vasa recta **passively preserve** it via countercurrent exchange.
- **Countercurrent Multiplication:**
  - *Descending Limb:* Water-permeable (AQP1) and solute-impermeable; tubular fluid is concentrated by osmotic water loss.
  - *Thick Ascending Limb (TAL):* Water-impermeable; actively pumps NaCl into the interstitium via NKCC2 ("the diluting segment"), delivering hypoosmotic fluid ($50–100$ mOsm/kg) to the distal nephron.
- **The Single Effect Multiplied:** A 200 mOsm/kgactive horizontal gradient across the ascending limb is multiplied by countercurrent flow into an axial gradient of over 1000 mOsm/kg.

Learning Objectives

1. Correlate the microanatomical structure of the nephron and its portal microcirculation with renal plasma flow and the fundamental mass balance equation.
2. Evaluate how the glomerular filtration barrier, Starling forces, and intrinsic autoregulatory mechanisms determine net filtration pressure and glomerular filtration rate.
3. Characterize the general epithelial transport mechanisms of proximal tubular reabsorption and peritubular secretion, highlighting their opposing roles in determining net excretion.
4. Describe how differing transport properties of the descending and ascending limbs of the Loop of Henle establish the hyperosmotic medullary gradient.
5. Explain how final urinary excretion and concentration are modulated in the collecting duct by the presence or absence of ADH and aldosterone.

Excretion & ADH: Dilute vs. Concentrated Urine

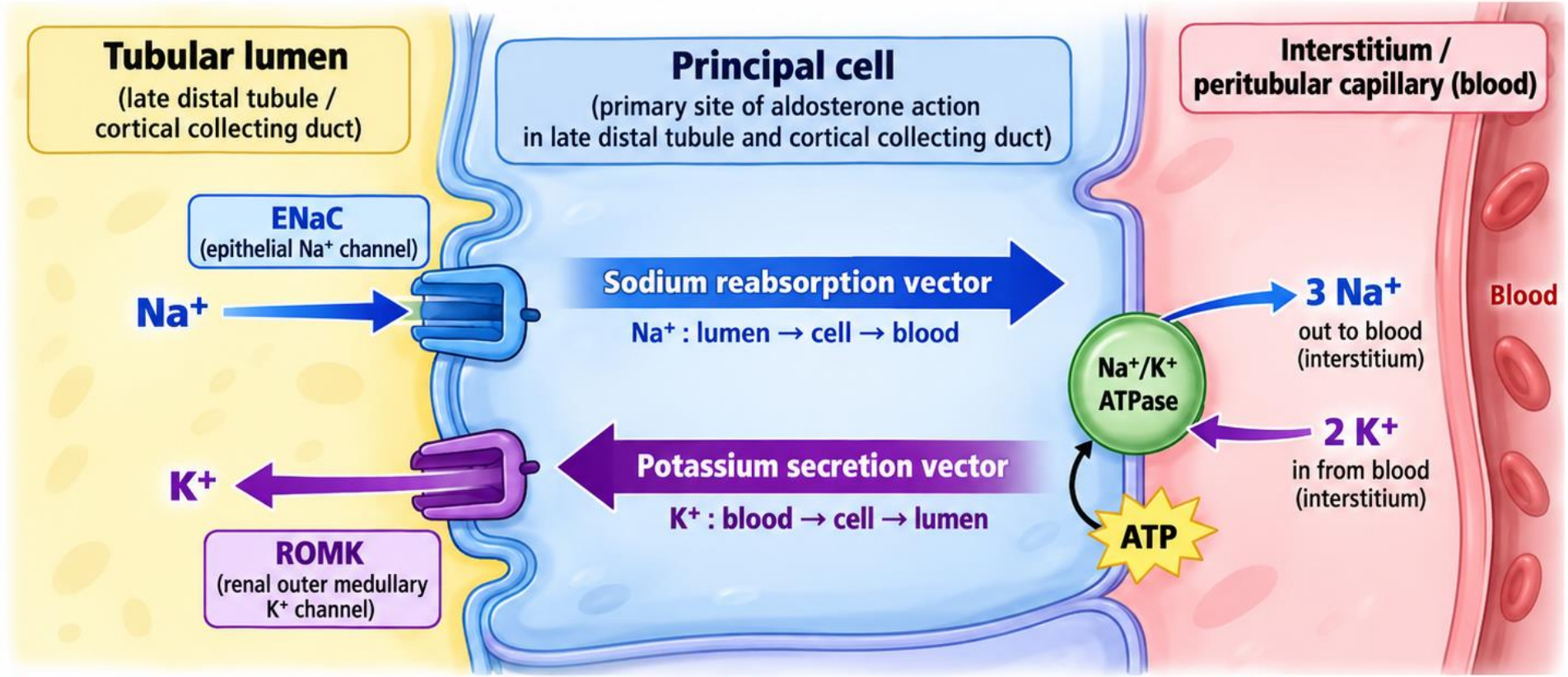
- **The Collecting Duct as the Final Arbiter:** While the loop of Henle *creates* the hyperosmotic medullary gradient, the collecting duct *uses* it to determine final urine concentration.
- **Absence of ADH (Hydrated State / Water Diuresis):**
  - Collecting duct luminal membrane remains **impermeable to water**.
  - Luminal fluid cannot equilibrate with the hypertonic interstitium.
  - Excretion: High volume of dilute, hypoosmotic urine (~ 50 mOsm/kg).
- **Presence of ADH (Dehydrated State / Antidiuresis):**
  - ADH inserts water channels (Aquaporin-2) into the luminal membrane.
  - Tubular fluid osmotically equilibrates with the hypertonic medulla as it descends.
  - Excretion: Low volume of concentrated, hyperosmotic urine (up to $1200–1400$ mOsm/kg).



ADH does not create the medullary gradient; it allows the collecting duct to use that gradient to reabsorb water.

Aldosterone: Fine-Tuning Final Sodium & Potassium Excretion

- **Fine-Tuning Solute Excretion:** While the proximal tubule reabsorbs the bulk of filtered sodium ($\sim 67\%$), aldosterone modulates the final $2–3\%$ to maintain hemodynamic stability.
- **Dual Transport Action:**
  - **Promotes** Na^+ **Reabsorption:** Increases luminal sodium entry channels and basolateral extrusion pumps, pulling sodium back into the blood.
  - **Promotes** K^+ **Secretion:** Drives potassium excretion into the tubular lumen to maintain electrolyte neutrality.
- **Primary Physiological Stimuli:**
  - Reduced renal perfusion / low blood volume (via Angiotensin II).
  - Hyperkalemia (direct adrenal stimulation).
- **Net Excretion Impact:** Aldosterone minimizes urinary Na^+ loss, expands extracellular fluid volume (water follows sodium when ADH is present), and eliminates excess dietary potassium.



Systemic stimuli for aldosterone release

Decreased effective circulating volume / decreased renal perfusion → RAAS activation

Elevated plasma K^+ → Aldosterone release

→ Aldosterone

Effects of aldosterone on principal cell

Aldosterone

- ↑ ENaC expression / activity (apical membrane)
- ↑ ROMK expression / activity (apical membrane)
- ↑ Na^+/K^+ ATPase expression / activity (basolateral membrane)

Physiologic result

Aldosterone promotes Na^+ reabsorption and K^+ secretion.

- ↑ Na^+ reabsorption (lumen → blood)
- ↑ K^+ secretion (blood → lumen)

Summary: ADH and Aldosterone

Feature	Antidiuretic Hormone (ADH)	Aldosterone
Primary Stimulus	Hyperosmolality; severe hypovolemia	Angiotensin II (low ECF volume); high [K^+]
Nephron Target	Entire collecting duct (medullary & cortical)	Late distal tubule & cortical collecting duct
Primary Substance Regulated	Water (free water reabsorption)	Solute (Na^+ reabsorption, K^+ secretion)
Effect on Final Urine	Concentrates urine (50 → 1200 mOsm/kg)	Reduces Na^+ excretion; increases K^+ excretion

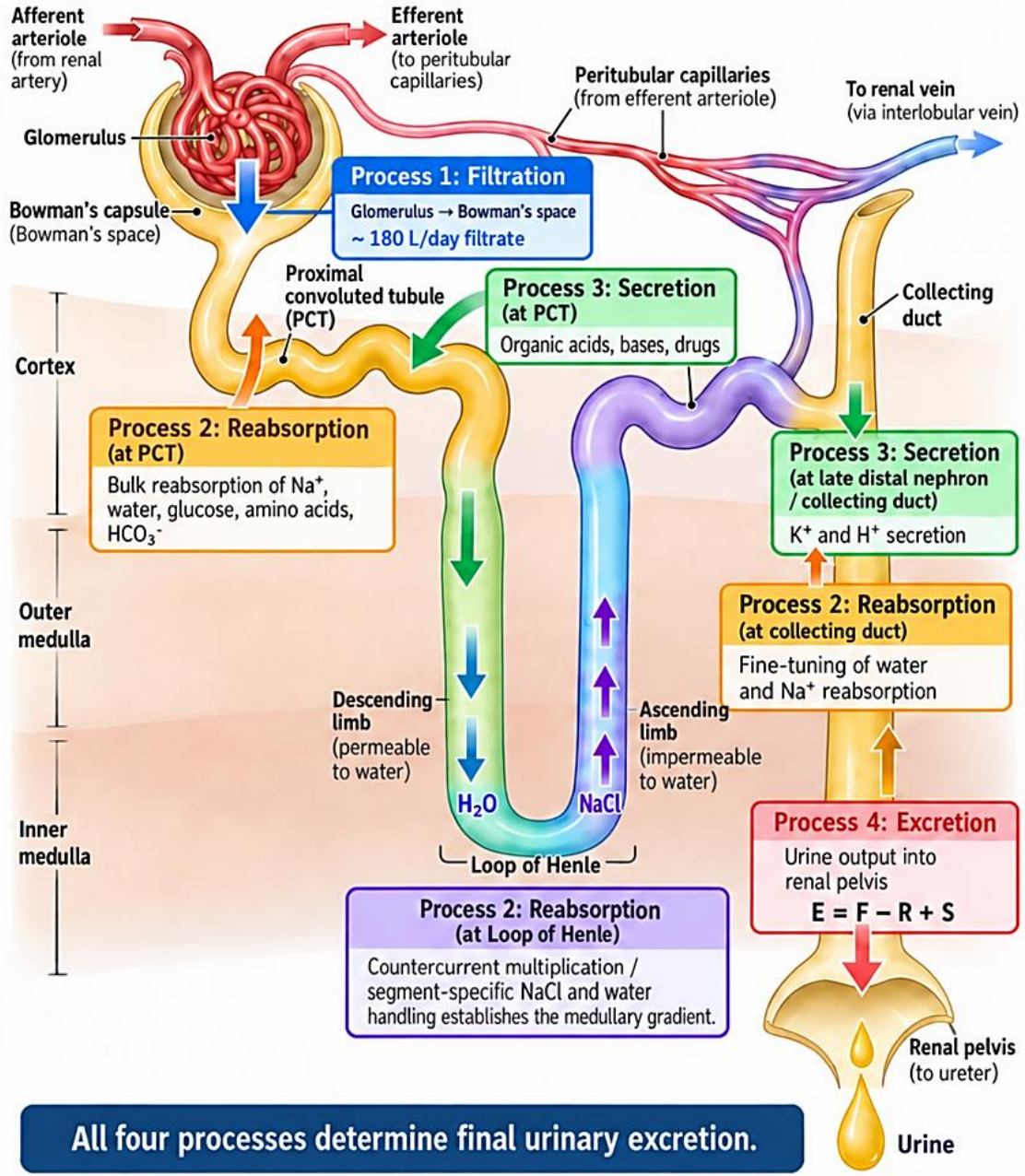
- **Takeaway:** Urinary excretion is not a single passive event; it is the calculated algebraic sum of filtration across a selective sieve, bulk and fine-tuned reabsorption, and targeted secretion.

Interim Summary 5

- **ADH is the Water Permeability Switch:** Secreted in response to hyperosmolality or hypovolemia, ADH inserts apical AQP2 channels in the collecting duct, allowing luminal fluid to equilibrate with the hyperosmotic medulla to produce concentrated urine ($1200–1400$ mOsm/kg). Without ADH, the duct remains watertight, producing dilute urine ($50–100$ mOsm/kg).
- **Aldosterone is the Salt & Potassium Fine-Tuner:** Stimulated by Angiotensin II and hyperkalemia, aldosterone acts on principal cells to upregulate apical ENaC and basolateral Na^+/K^+-ATPase (reabsorbing the final $2–3\%$of Na^+) while promoting K^+secretion via ROMK.
- **Core Takeaway:** Filtration creates the raw material, the PCT and Loop handle bulk processing, and the Collecting Duct delivers homeostatic precision.

THE NEPHRON: MASTER SUMMARY

Anatomy, major processes, and key functions along the nephron



All four processes determine final urinary excretion.

ADH vs. Aldosterone

	ADH (Antidiuretic hormone, Vasopressin)	Aldosterone
Primary stimulus	↑ plasma osmolality; significant ↓ effective circulating volume	RAAS activation (Ang II) and ↑ plasma K^+
Source	Synthesized in hypothalamus; released from posterior pituitary	Zona glomerulosa of adrenal cortex
Main target nephron site	Late distal tubule and collecting duct	Principal cells of late distal tubule and cortical collecting duct
Main transport effect	Inserts AQP2 in apical membrane → ↑ water permeability	↑ ENaC, ↑ Na^+/K^+ ATPase, ↑ ROMK → ↑ Na^+ reabsorption and ↑ K^+ secretion
Main physiologic role	Water conservation; concentrates urine	ECF volume support and K^+ homeostasis
Effect on urine volume/ osmolality	↓ urine volume, ↑ urine osmolality	Promotes Na^+ retention; indirectly reduces Na^+ loss, not the primary hormone controlling urine osmolality

ADH primarily regulates water permeability; aldosterone primarily regulates Na^+ reabsorption and K^+ secretion.

The End